

Computer Vision 1

Retrieval, Detection, Segmentation

Dr. Martin R. Oswald

Lecture overview

Week	Date	Presenter	Lectures	Exercise / Labs
1	36	02/09/2024	Arnoud Visser	1. Introduction and Course Logistics; Camera model
1	36	05/09/2024	Arnoud Visser	2. Sensor and image formation Color
2	37	09/09/2024	Arnoud Visser	3. In place image processing Photometric stereo
2	37	12/09/2024	Arnoud Visser	4. Convolution and image filtering
3	38	16/09/2024	Arnoud Visser	5. Feature: edge and corner
3	38	19/09/2024	Arnoud Visser	6. Motion and optical flow
4	39	23/09/2024	Martin Oswald	7. Global transform, Image stitching, alignment
4	39	26/09/2024	Martin Oswald	8. Basics of Recognition 1
5	40	30/09/2024	Martin Oswald	9. Basics of Recognition 2
5	40	03/10/2024	Martin Oswald	10. Convolutional neural network 1 (Basic)
6	41	07/10/2024	Martin Oswald	11. Convolutional neural network (Training and testing)
6	41	10/10/2024	Martin Oswald	12. Stereo, 3D reconstruction
7	42	14/10/2024	TBA	13. Guest lectures
7	42	17/10/2024	TBA; Martin & Arnoud	14. Guest lecture; Course Wrap-up, Exam Q&A
8	43	21/10/2024	Your turn!	Final exam

Today

Online

Lectures

Recap

How many features are needed?

The Bag of Words Representation

I love this movie! It's sweet, but with satirical humor. The dialogue is great and the adventure scenes are fun... It manages to be whimsical and romantic while laughing at the conventions of the fairy tale genre. I would recommend it to just about anyone. I've seen it several times, and I'm always happy to see it again whenever I have a friend who hasn't seen it yet!



it	6
I	5
the	4
to	3
and	3
seen	2
yet	1
would	1
whimsical	1
times	1
sweet	1
satirical	1
adventure	1
genre	1
fairy	1
humor	1
have	1
great	1
...	...

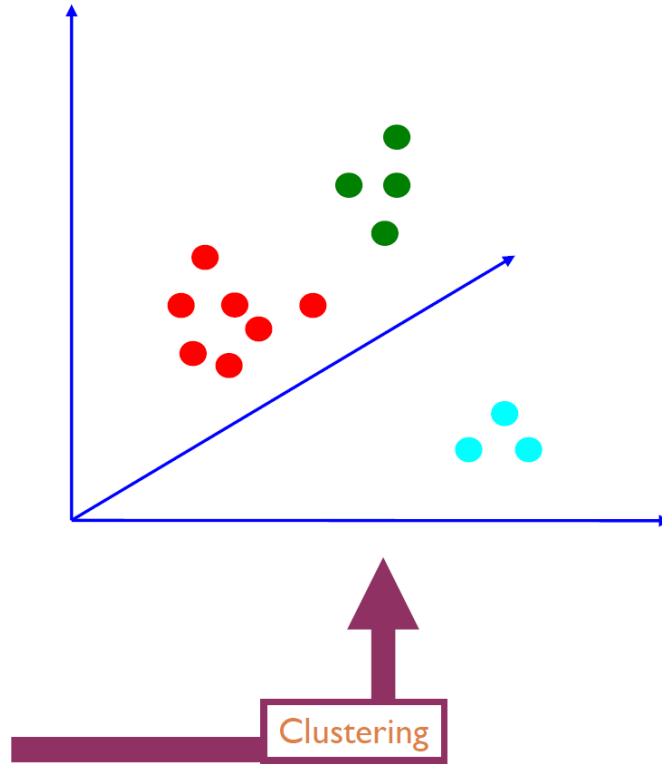
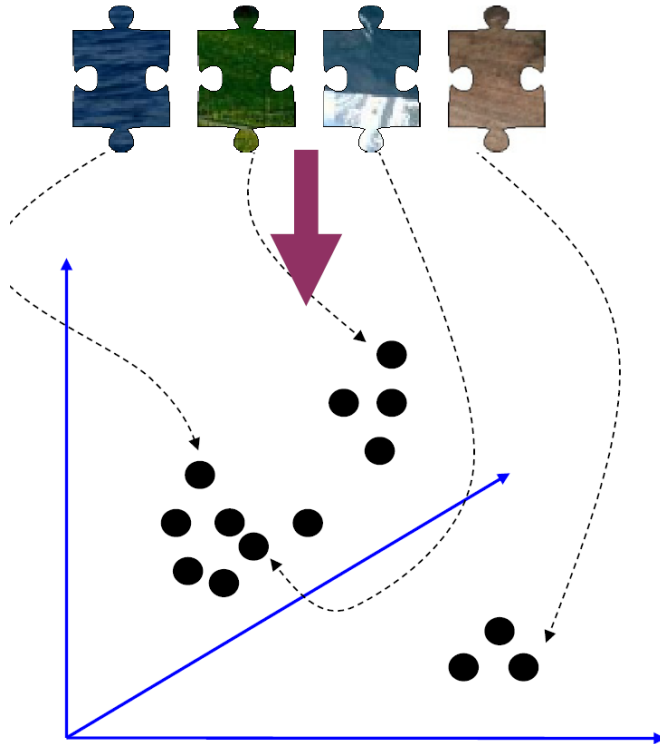
Bag of Visual Words



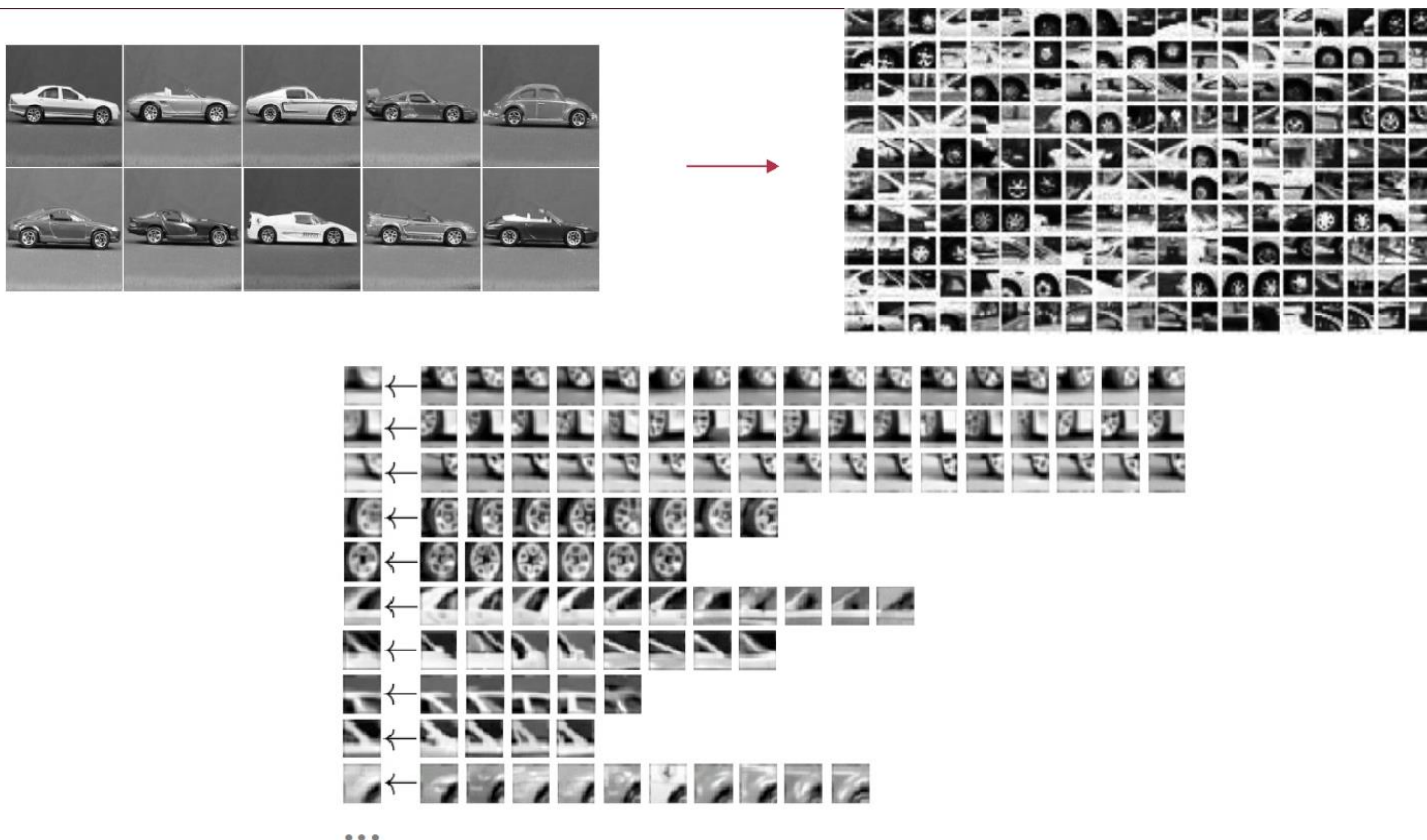
Slide credits: Jan van Gemert

Feature Clustering

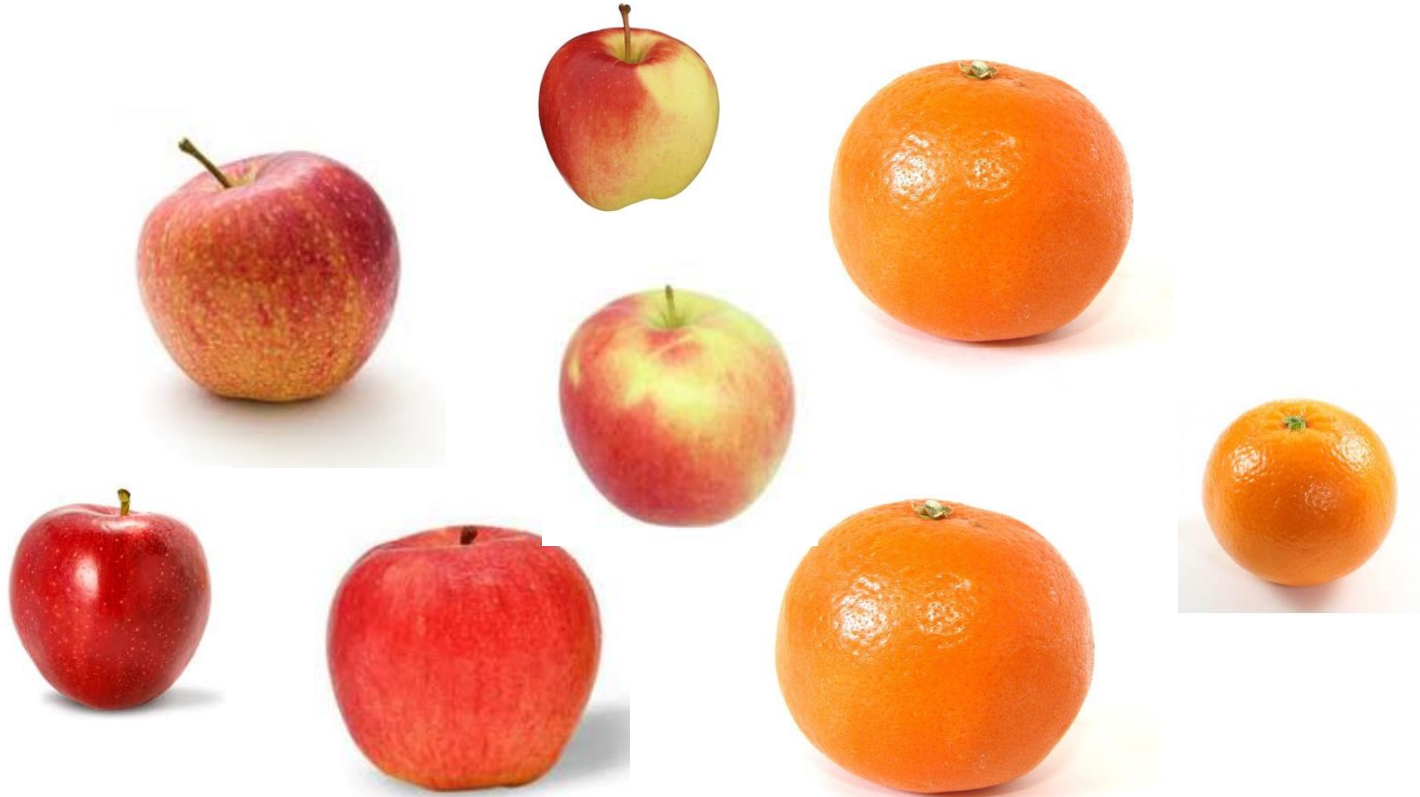
CLUSTERING FOR BOW



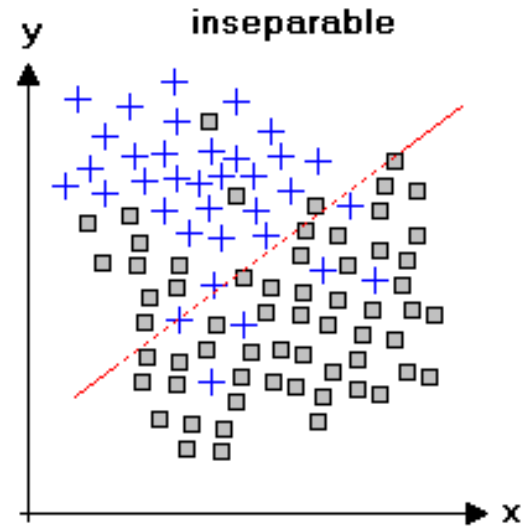
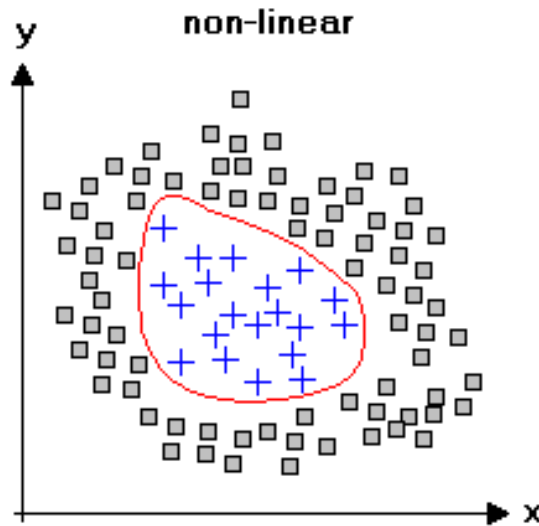
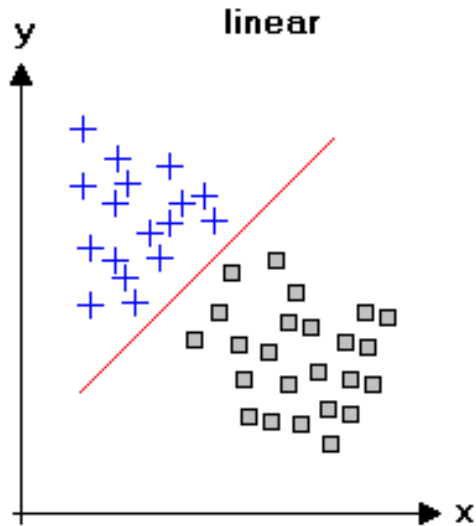
Codebook Example



Apple - Orange Classification



Feature Spaces - Separability



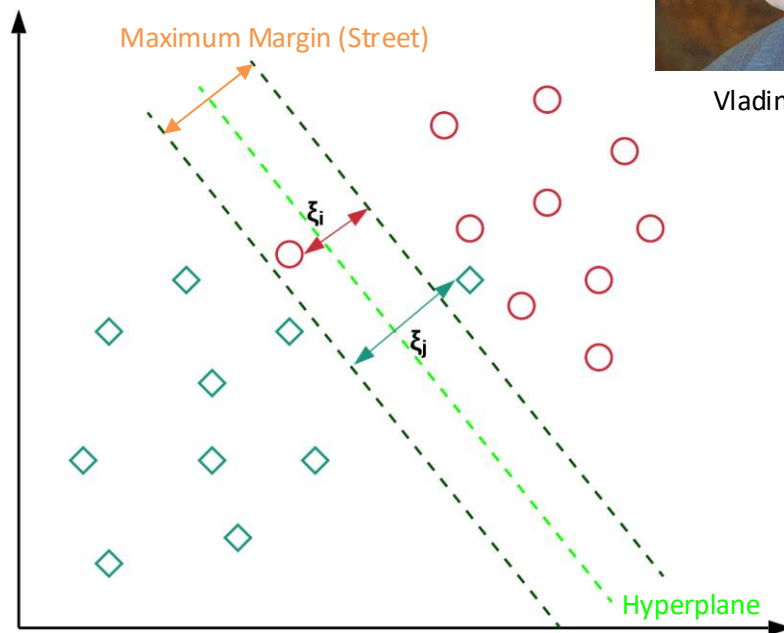
Support Vector Machines (SVMs)

- Goal: Find separating hyperplane (1D: point, 2D: line, 3D: plane, etc.)
- Maximizes margin (street) between classes
- Support vectors are data points which are closest to the decision boundary
- Only support vectors define the decision boundary (no other data points)
- Support vectors are found via optimization if linear separation exists (**Hard Margin SVM**)
- **Soft Margin SVMs** allow for outliers by additionally minimizing slack variables ξ_i that shift separation boundary
- Extension to non-linear decision boundaries via kernel trick

SVMs work very well and are quite robust to noisy data!



Vladimir Vapnik



[Good explanation for kernel trick: <https://youtu.be/O7yT0-5VII>]

- Accuracy:

$$\text{Accuracy} = \frac{\text{TN} + \text{TP}}{\text{TN} + \text{FP} + \text{TP} + \text{FN}}$$

- Precision and Recall:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

- F1 score: geometric mean of precision and recall

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Precision and Recall

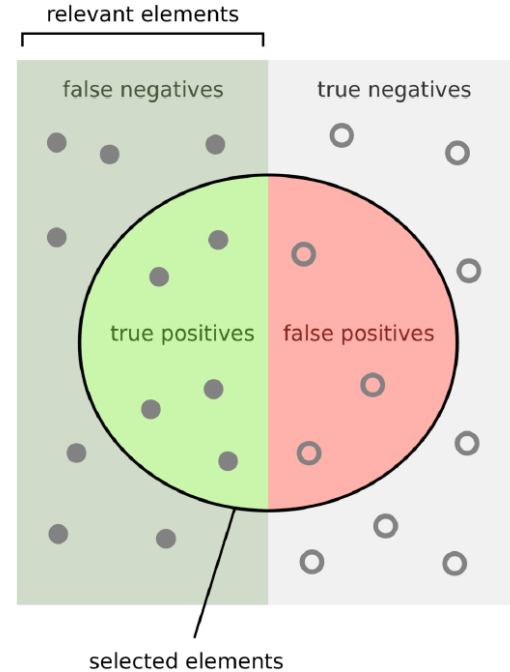
How many selected items are relevant?

$$\text{Precision} = \frac{\text{green semi-circle}}{\text{green and red semi-circle}}$$

How many relevant items are selected?

$$\text{Recall} = \frac{\text{green semi-circle}}{\text{green rectangle}}$$

Combine P+R into: F1-Measure



- Chapter 6.3
- Chapter 6.4 (introduction)
- Chapter 6.4.2

Image Understanding

AN IMAGE IS WORTH A THOUSAND WORDS



Image Understanding

Classification



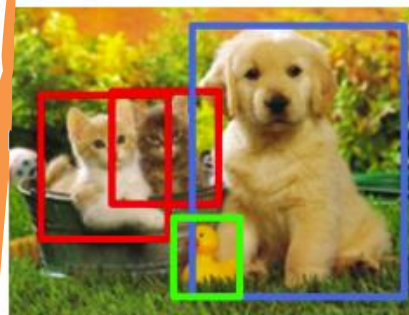
CAT

**Classification
+ Localization**



CAT

Object Detection



CAT, DOG, DUCK

**Instance
Segmentation**



CAT, DOG, DUCK

- Object Classification: classify the main object in an image if the image has one main object.
- Visual Image Retrieval: retrieve images from the library through a query image.
- Object Detection: identify whether there is an object in an image and recognize the object.
- Image Segmentation: segment an image into regions according the pixels the object belong to.

Today: Object Detection!

Today's Agenda



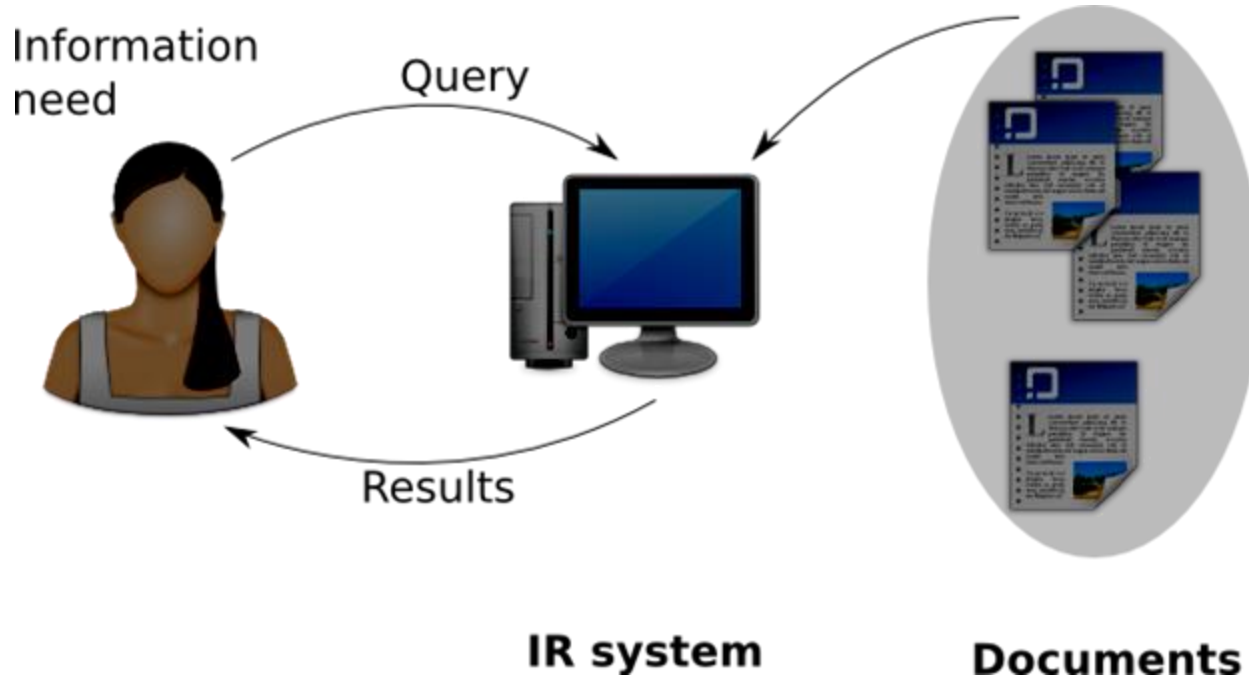
- **Image Retrieval**
- **Object Detection**
- **Image Segmentation**

Today's Agenda



- **Image Retrieval**
- Object Detection
- Image Segmentation

What is Image Retrieval?



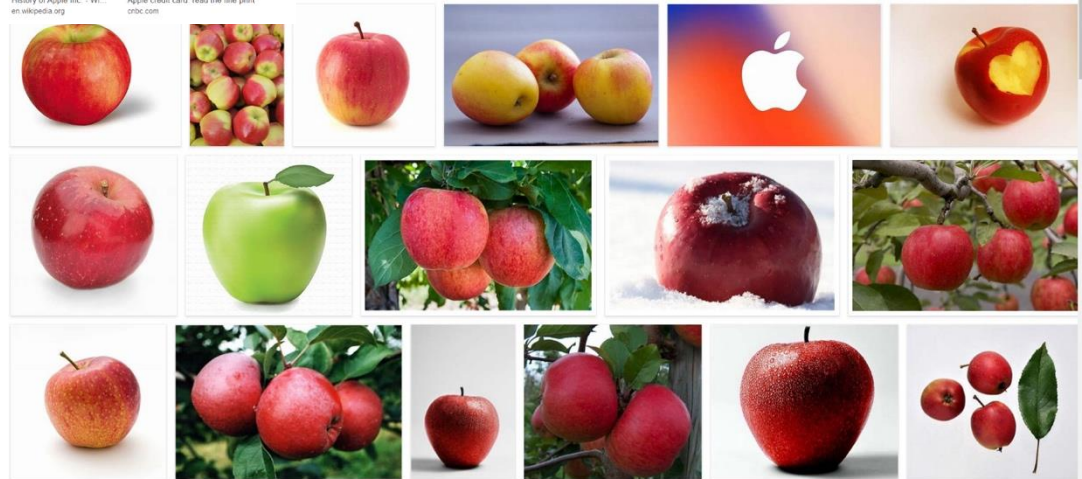
What is Image Retrieval?



Image Retrieval



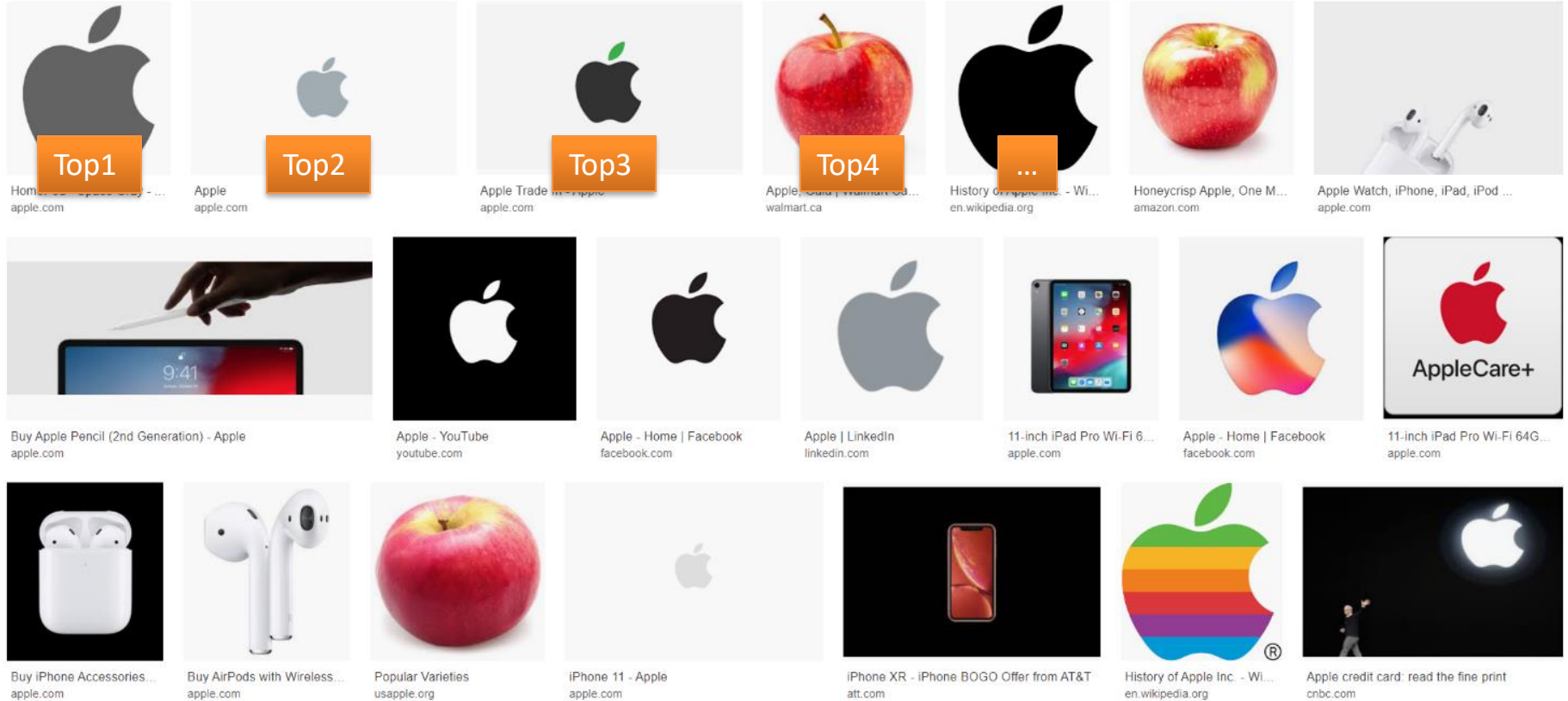
Google Search



Search term: "appel"

Bing Search

Image Retrieval



Google Search

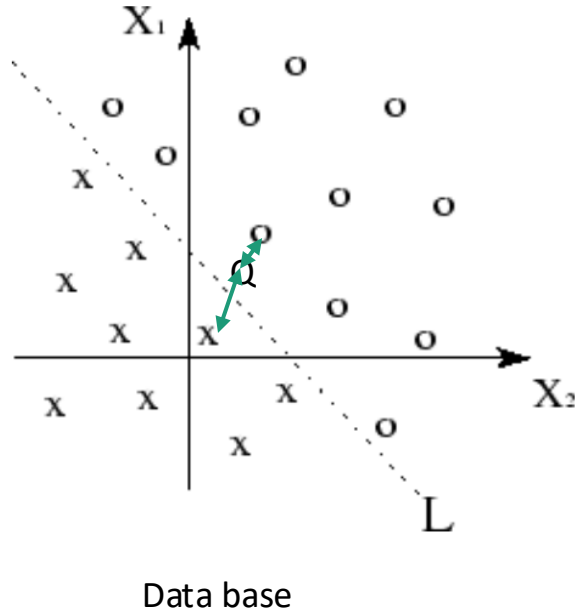
Bag of Words for Image Retrieval

- Offline: construct BoW Vocabulary
- Offline: All images in the dataset -> BoWs
- Create description from Query image
- Compare to all images in the dataset
- Rank according to similarity

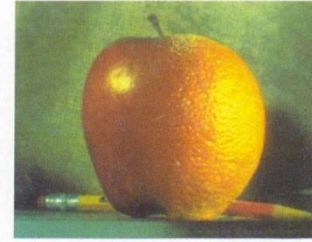


Slide credits: Jan van Gemert

Retrieval Example



1. A Query: Q



2. Extract feature

3. Get similarity

Top 1: o

Top 2: x

...

Retrieval Evaluation

Precision@K

- Set a rank threshold K
- Compute % relevant in top K
- Ignores images ranked lower than K

$$P@K = \frac{TP}{\underbrace{TP + FP}_K} = \frac{TP}{K}$$

- Example: 
 - Prec@3 of 2/3
 - Prec@4 of 2/4
 - Prec@5 of 3/5

Retrieval Evaluation

Average Precision@K

- Consider rank position of each relevant images
 - $K_1, K_2, \dots, K_R$

- Compute Precision@K for each $K_1, K_2, \dots, K_R$

- Average precision = average of P@K

- Example:  has average precision:

rel: 1 0 1 0 1

$$AP@K = \frac{1}{3} * \left(\frac{1}{1} * 1 + \frac{1}{2} * 0 + \frac{2}{3} * 1 + \frac{2}{4} * 0 + \frac{3}{5} * 1 \right) \approx 0.76$$

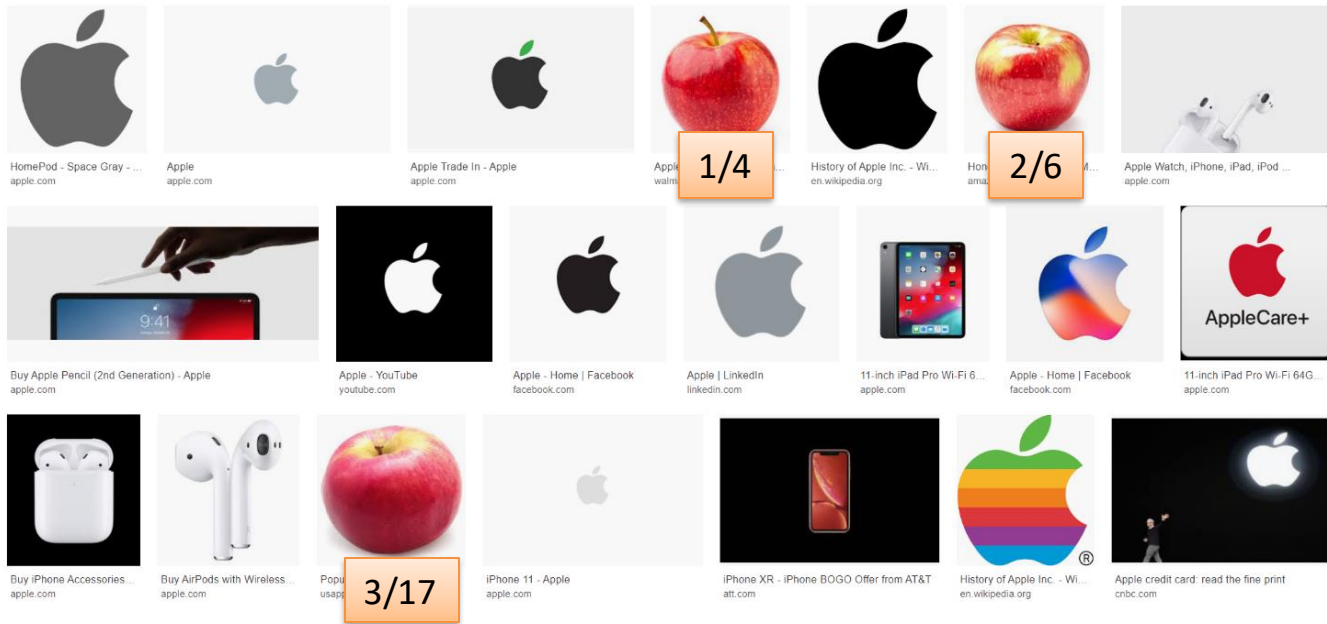
relevance: {0,1}

↓

$$AP@K = \frac{\sum_{k=1}^K (P@k * rel_k)}{\# \text{ relevant results}}$$

Retrieval Evaluation

P@K and AP@K for “apple” (relevant: food)



$$\text{Average Precision @ 17} = \frac{1}{3} * (\frac{1}{4} + \frac{2}{6} + \frac{3}{17})$$

Retrieval Evaluation

Mean Average Precision (mAP)

- Take the mean over multiple queries/rankings

$$MAP@K = \frac{1}{Q} \sum_{q=1}^Q AP@kq$$

- Example:

Ranking #1: $(1.0 + 0.67 + 0.75 + 0.8 + 0.83 + 0.6)/6 = 0.78$

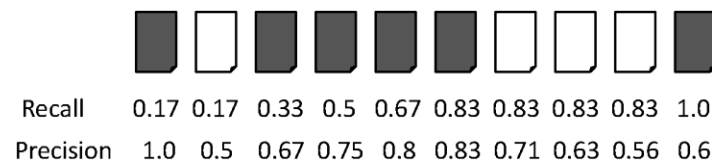
Ranking #2: $(0.5 + 0.4 + 0.5 + 0.57 + 0.56 + 0.6)/6 = 0.52$

$$MAP@K = \frac{1}{2} * (0.78 + 0.52)$$

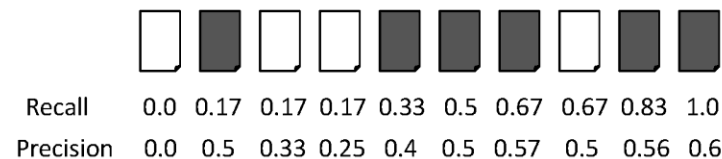


= the relevant documents

Ranking #1



Ranking #2



Today's Agenda



- Image Retrieval
- **Object Detection**
- Image Segmentation

Object Detection



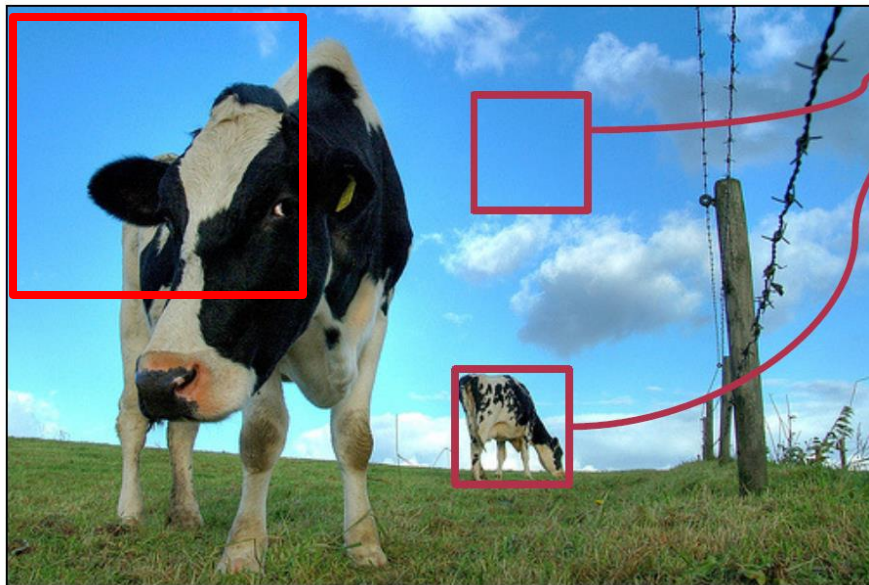
[<https://youtu.be/7BgiFrPs-x8?si>]

- **Sliding window**
- **Selective Search**
- **Region proposals (next lecture)**

How to Find The Bounding Box?

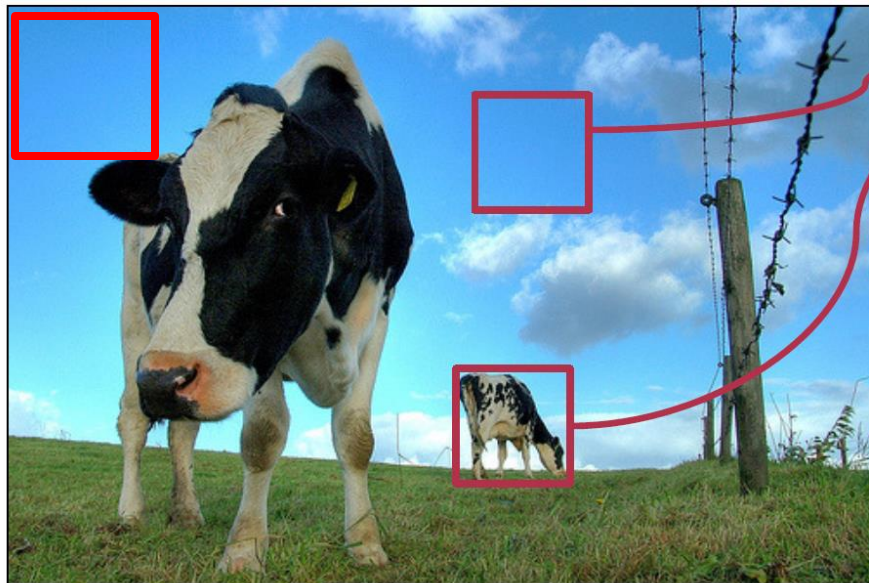


Sliding Window



Is this a cow?

Sliding Window



Is this a cow?

Sliding Window



- + Many boxes
- + Many scales
- + Many box ratios
- + Many classes
- = **Computational Expensive**

Selective Search

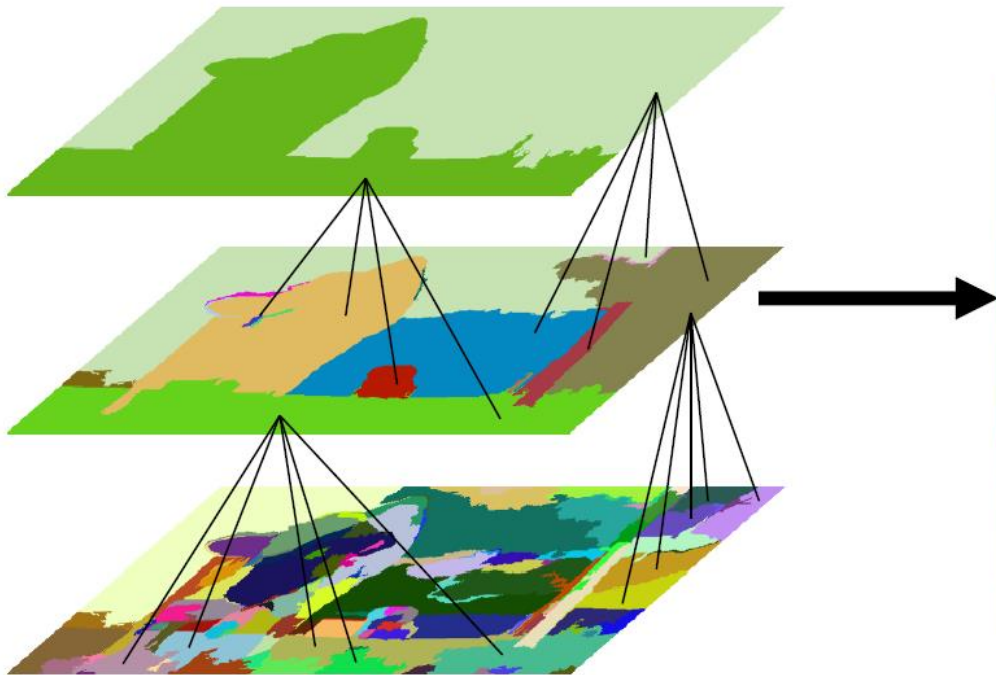
- Idea: Images are hierarchical



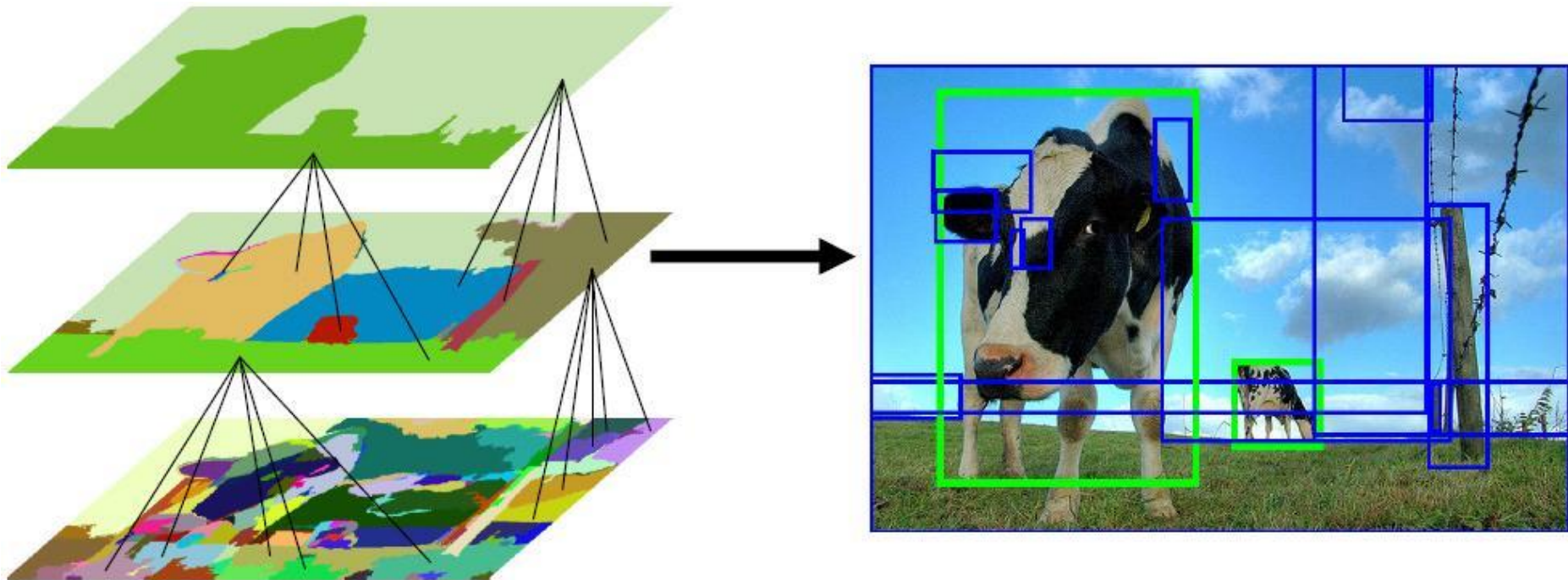
- Initial segments from over-segmentation



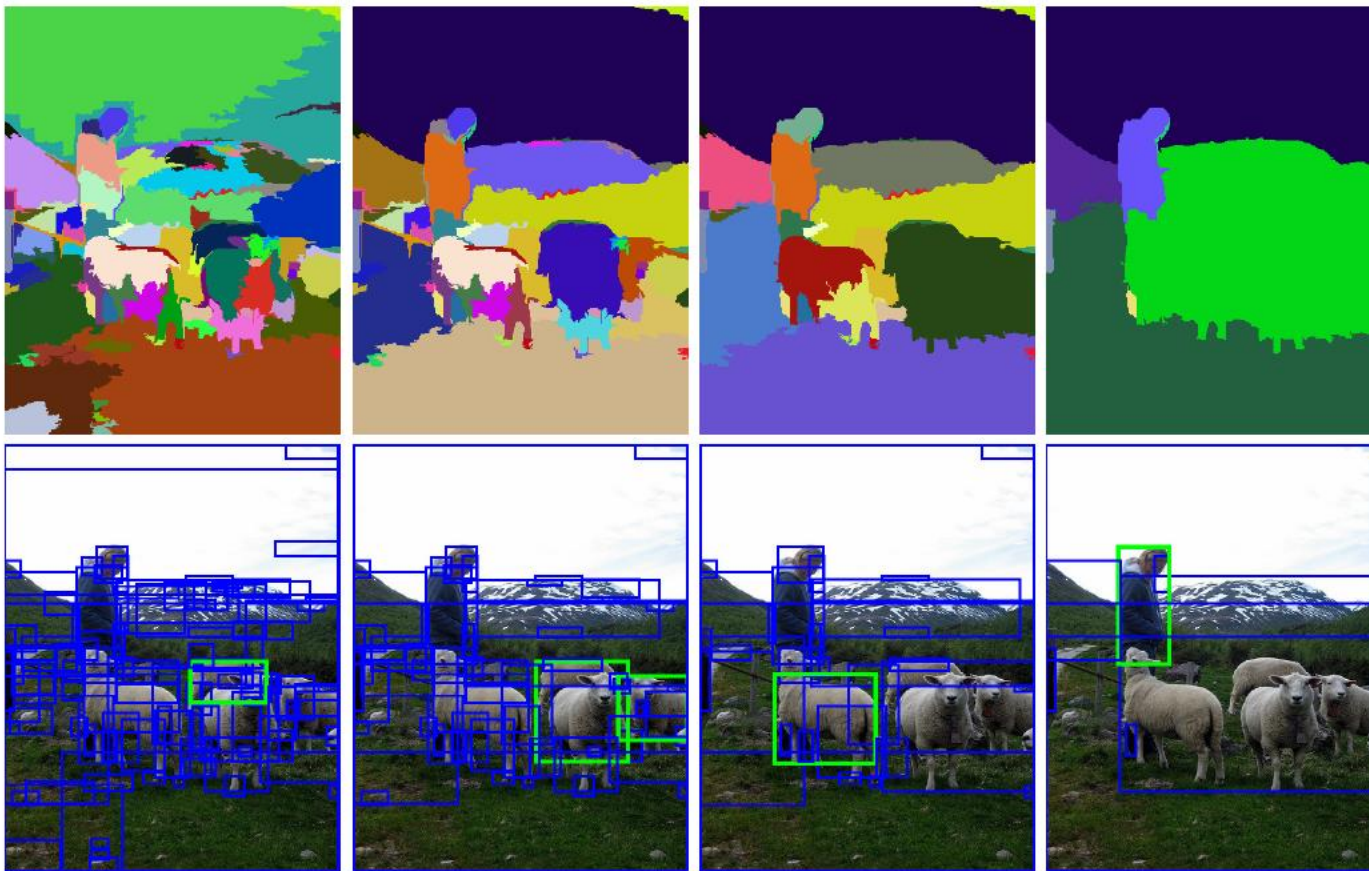
Selective Search



Selective Search



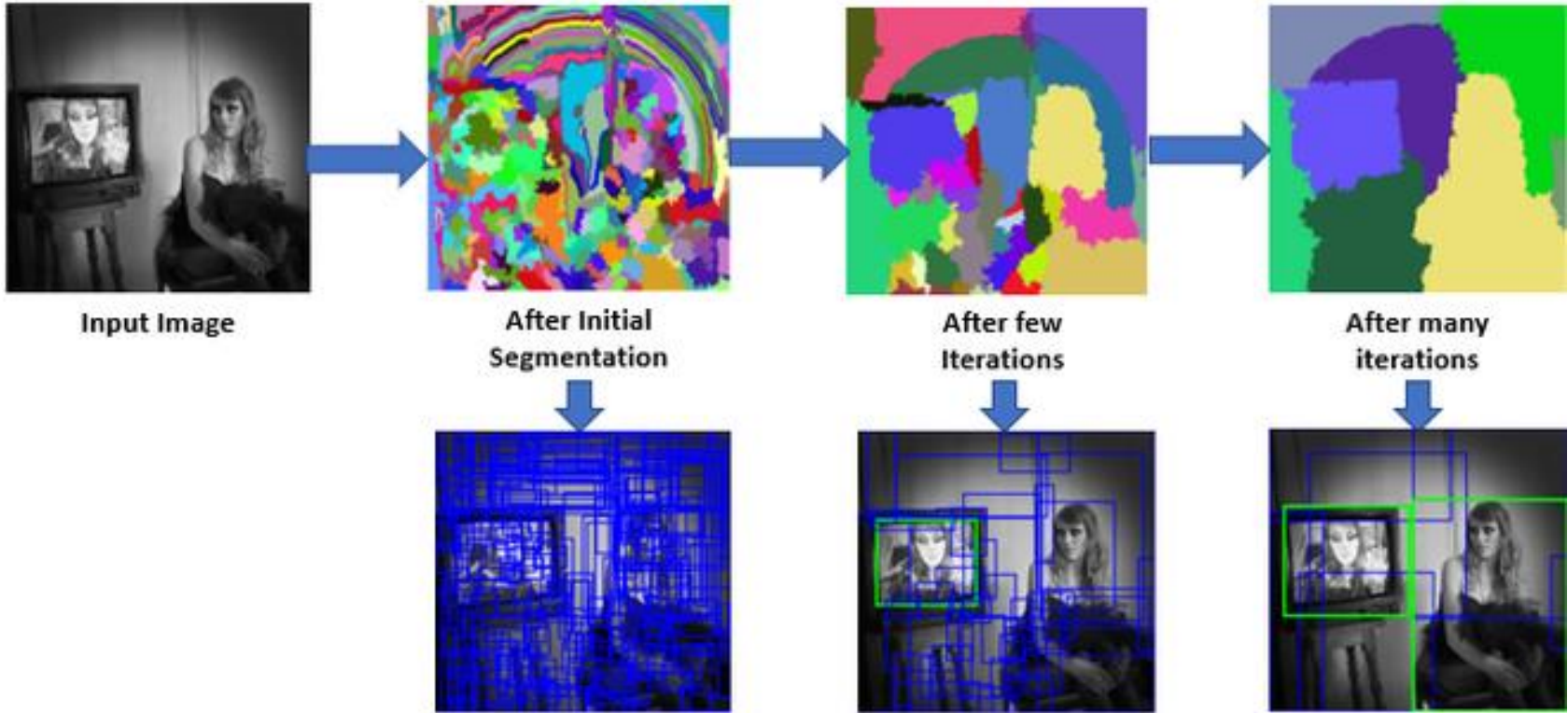
Selective Search - Examples



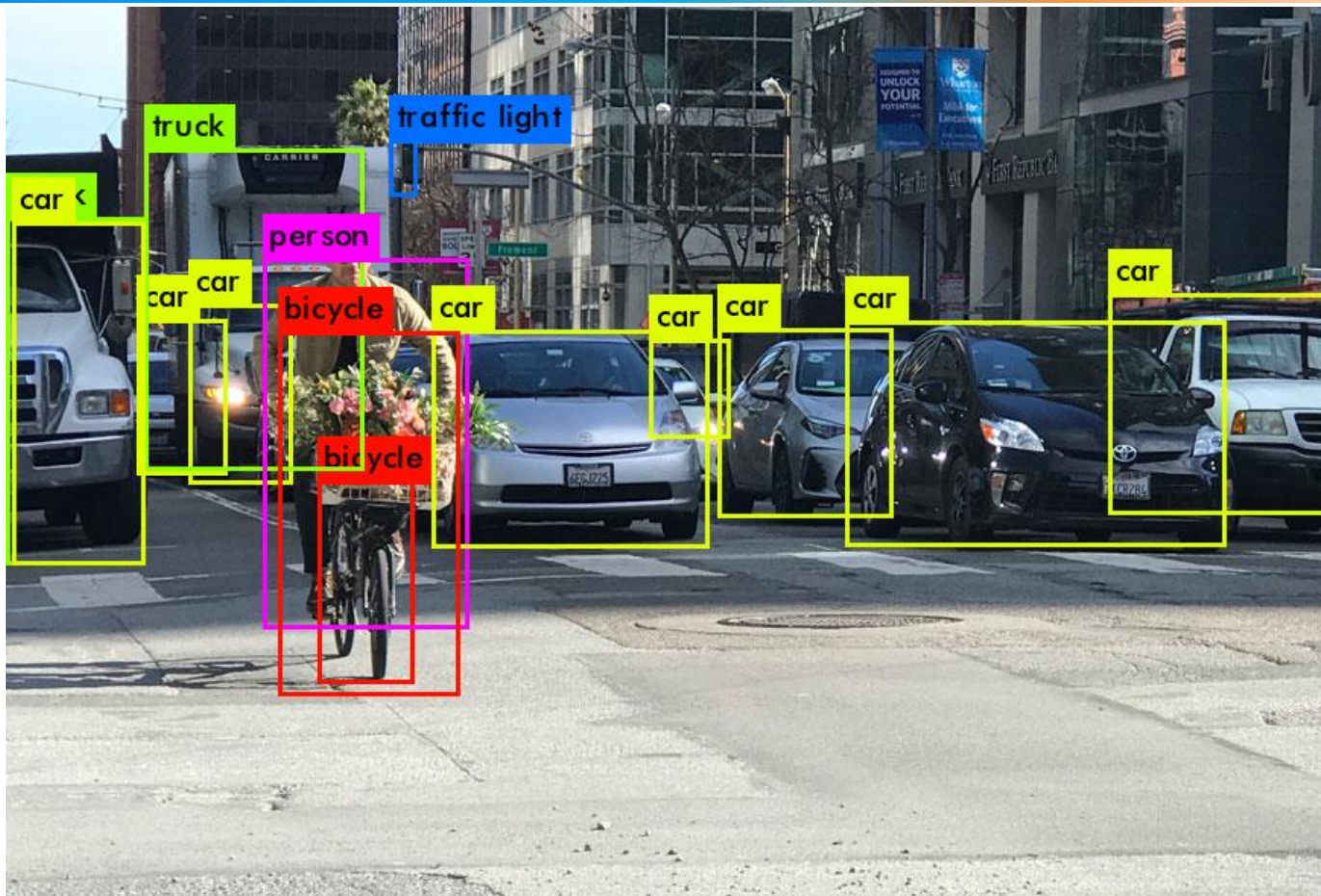
Selective Search - Examples



Selective Search - Examples



You only look once (YOLO)



You only look once (YOLO)

A graphic for YOLO v2. It features a dark teal background with a network of glowing nodes and connecting lines. The text 'YOLO v2' is prominently displayed in the center in a large, white, sans-serif font. At the bottom, a URL is provided in a smaller white font.

YOLO v2

<http://pureddie.com/yolo>

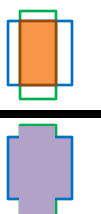
Autonomous Driving



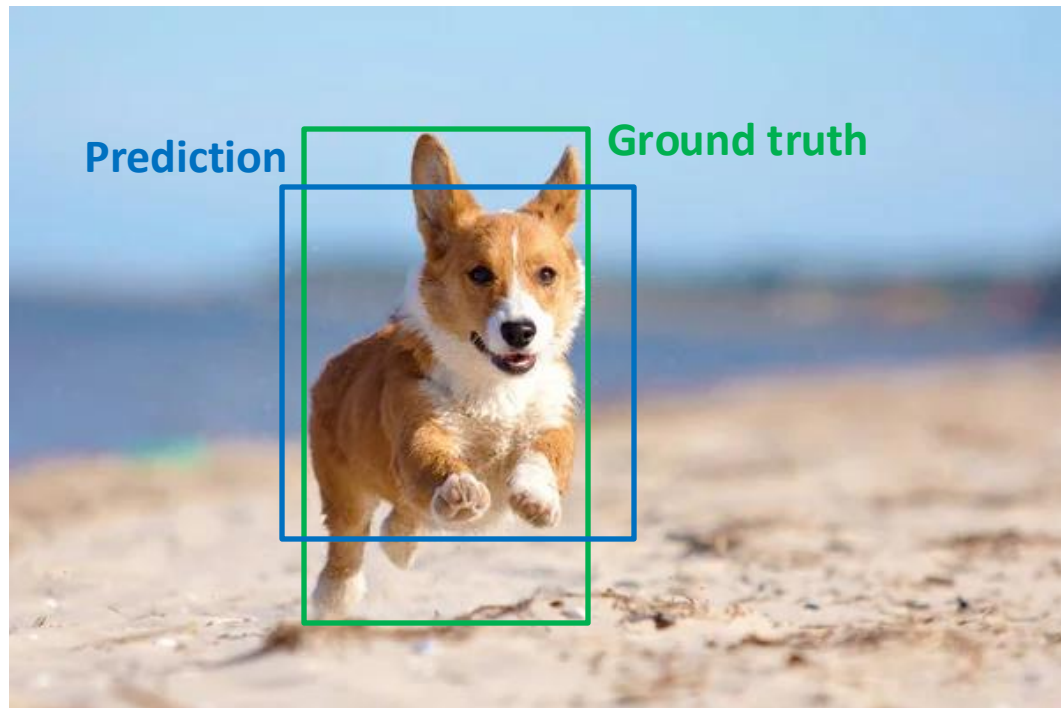
- How can we evaluate object detection?
- If we binary select inliners and outliers of predicted bounding boxes, we can use retrieval evaluation measures:
Precision and Recall, AP, mAP, etc.

Comparing Bounding Boxes

- Intersection over Union (IoU)
(a.k.a. “Jaccard similarity/index”)

$$\text{IoU} = \frac{\text{Intersection}}{\text{Union}}$$


- Typically choose a threshold
e.g. $\text{IoU} \geq 0.5$, or 0.7
between prediction and GT
to decide about correct or
incorrect predictions



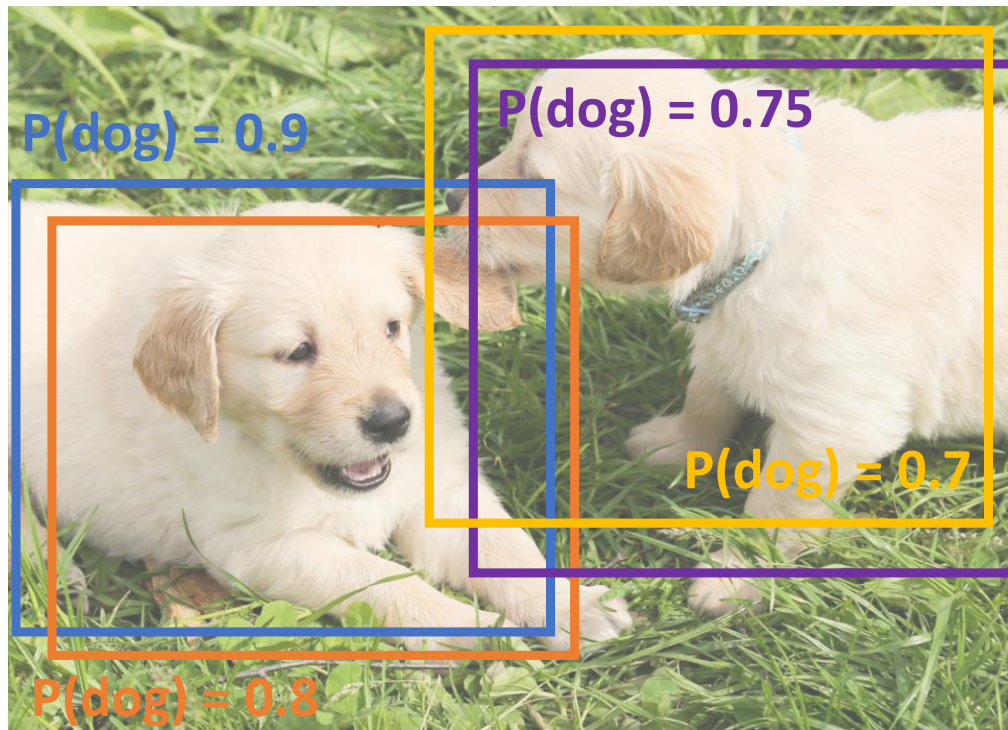
[<https://depositphotos.com/photos/corgi-running.html>]

Duplicate Filter: Non-Maximum Suppression (NMS)

Problem: Object detectors often output many overlapping detections

Solution: Post-process raw detections using **Non-Max Suppression (NMS)**:

1. Select next highest-scoring box
2. Eliminate lower-scoring boxes with $\text{IoU} > \text{threshold}$ (e.g. 0.7)
3. If any boxes remain, GOTO 1



[Puppy image is [CC0 Public Domain](#)]

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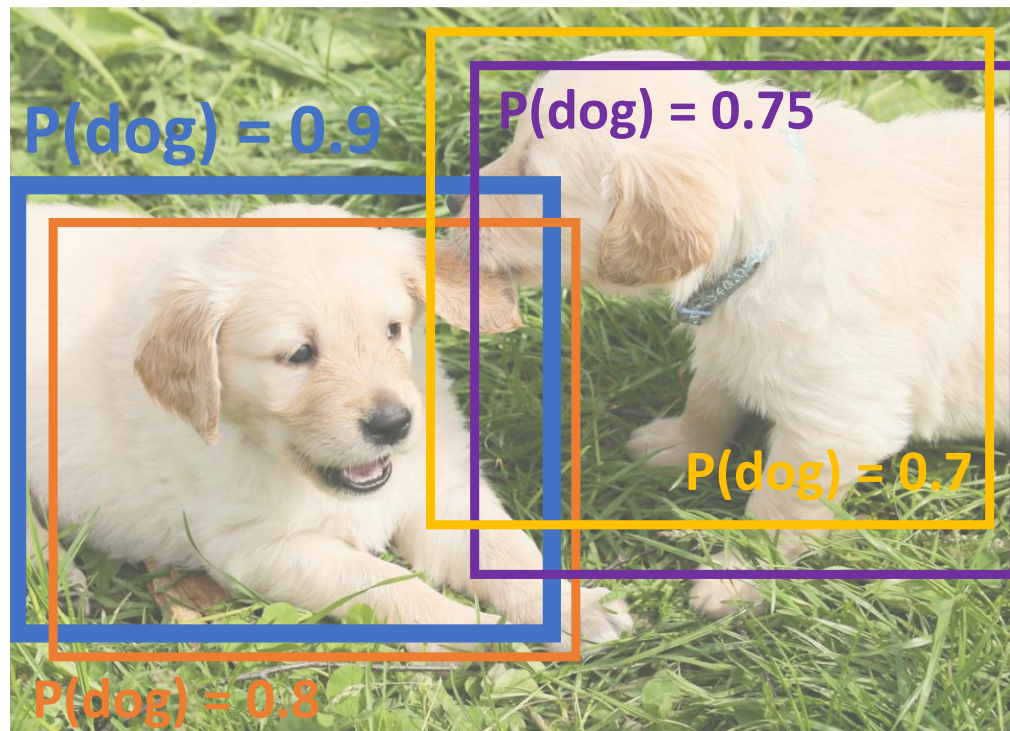
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$$\text{IoU}(\blacksquare, \blacksquare) = \mathbf{0.78}$$

$$\text{IoU}(\blacksquare, \blacksquare) = 0.05$$

$$\text{IoU}(\blacksquare, \blacksquare) = 0.07$$



[Puppy image is [CC0 Public Domain](#)]

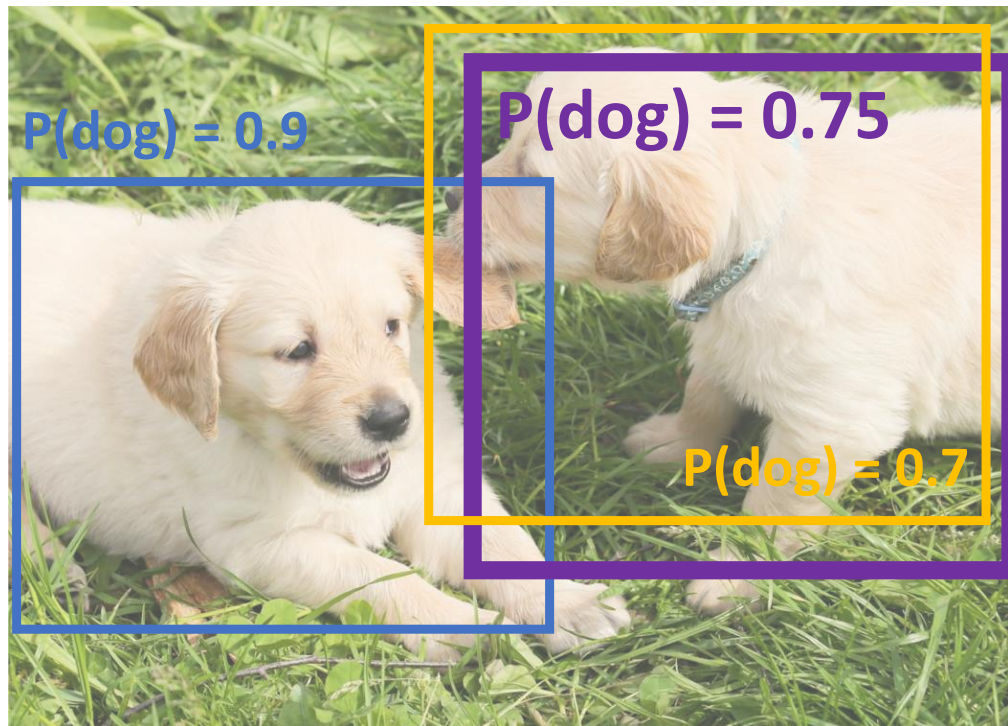
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$$\text{IoU}(\blacksquare, \blacksquare) = \mathbf{0.74}$$



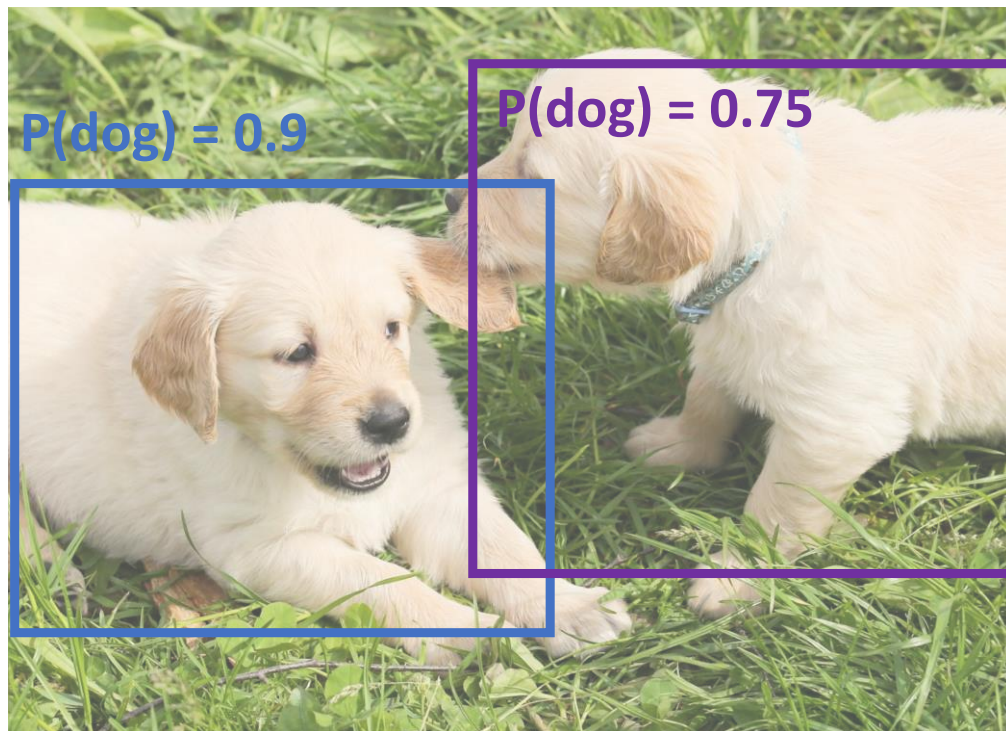
[Puppy image is [CC0 Public Domain](#)]

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[Crowd image is for commercial use under the [Pixabay license](#)]

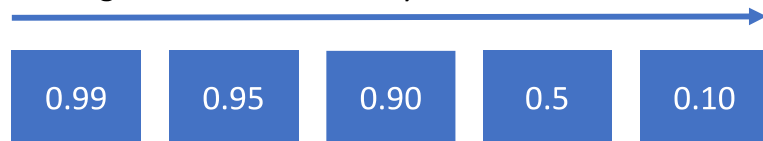
Problem: NMS may eliminate “good” boxes when objects are highly overlapping. There is not yet a good solution.

Evaluating Object Detectors

Mean Average Precision (mAP):

1. Run object detector on all test images (with NMS)
2. For each category, compute Average Precision (AP) = area under Precision vs Recall Curve
 1. For each detection (highest score to lowest score)

All dog detections sorted by score

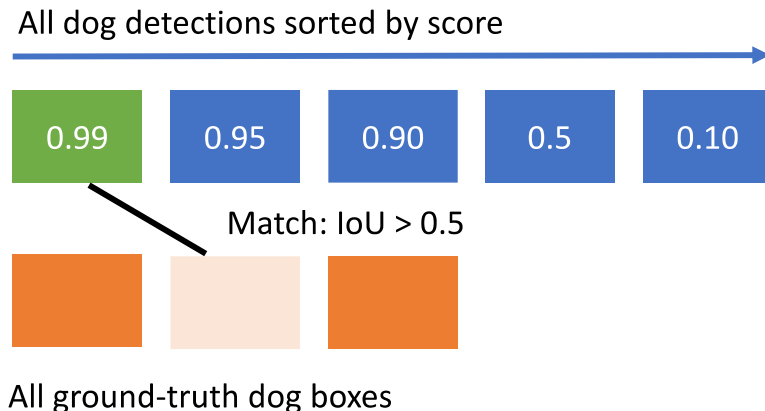


All ground-truth dog boxes

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 1. If it matches some GT box with $\text{IoU} > 0.5$, mark it as positive and eliminate the GT
 2. Otherwise mark it as negative

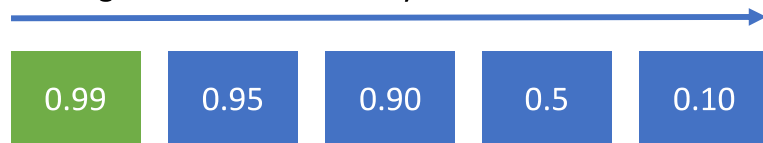


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 3. Plot a point on PR Curve

All dog detections sorted by score



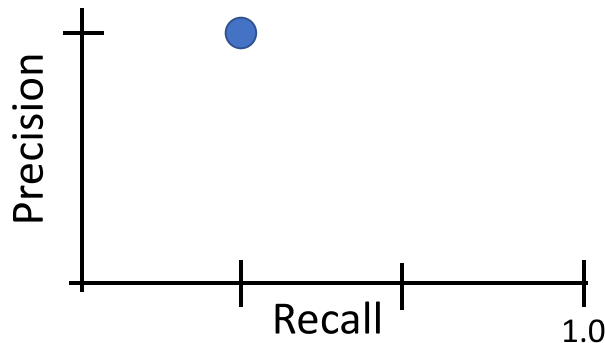
Match: IoU > 0.5



All ground-truth dog boxes

$$\text{Precision} = 1/1 = 1.0$$

$$\text{Recall} = 1/3 = 0.33$$

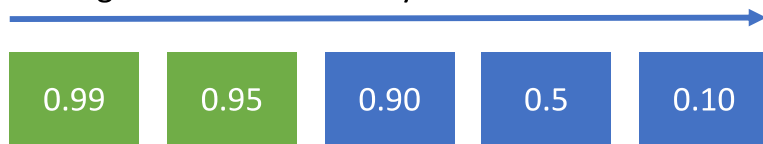


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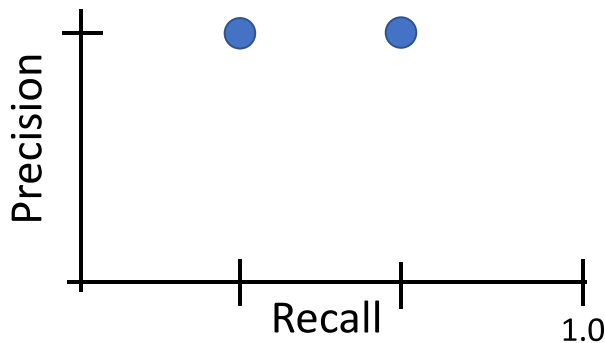
Match: IoU > 0.5



All ground-truth dog boxes

$$\text{Precision} = 2/2 = 1.0$$

$$\text{Recall} = 2/3 = 0.67$$

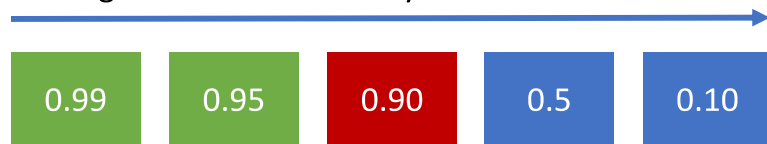


Evaluating Object Detectors

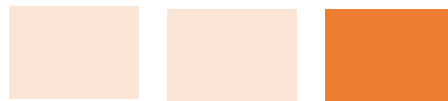
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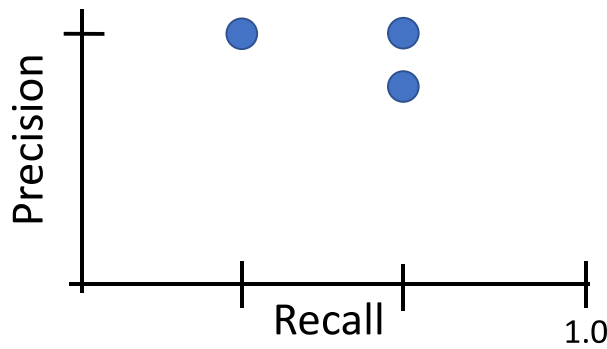
No match > 0.5 IoU with GT



All ground-truth dog boxes

$$\text{Precision} = 2/3 = 0.67$$

$$\text{Recall} = 2/3 = 0.67$$

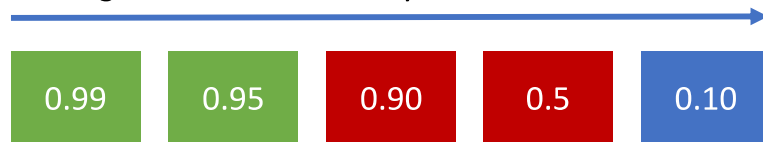


Evaluating Object Detectors

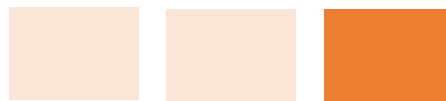
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All dog detections sorted by score



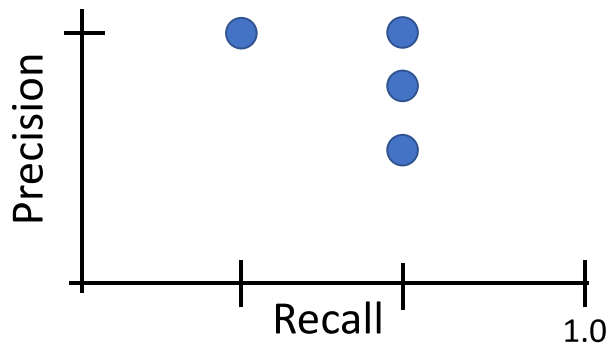
No match > 0.5 IoU with GT



All ground-truth dog boxes

$$\text{Precision} = 2/4 = 0.5$$

$$\text{Recall} = 2/3 = 0.67$$

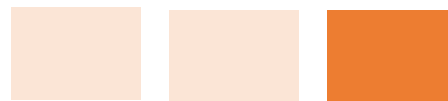
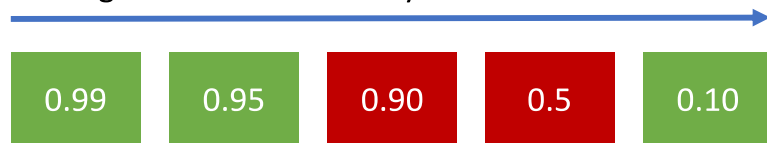


Evaluating Object Detectors

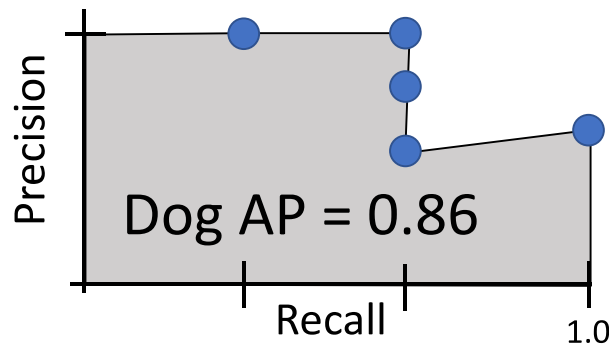
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All dog detections sorted by score



All ground-truth dog boxes

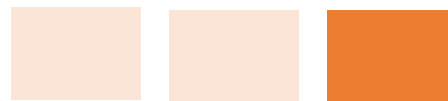
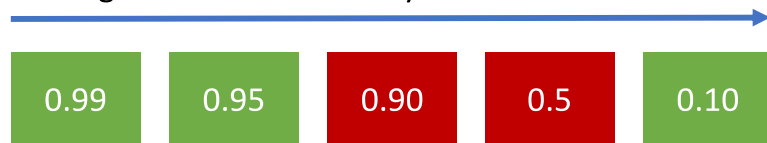


Evaluating Object Detectors

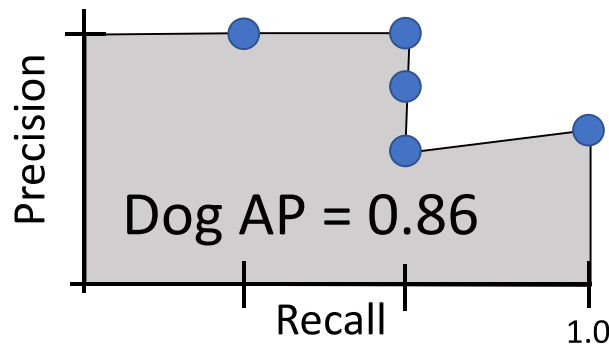
Mean Average Precision (mAP):

1. Run object detector on all test images (with NMS)
2. For each category, compute Average Precision (AP) = area under Precision vs Recall Curve
 1. For each detection (highest score to lowest score)
 1. If it matches some GT box with IoU > 0.5, mark it as positive and eliminate the GT
 2. Otherwise mark it as negative
 3. Plot a point on PR Curve

All dog detections sorted by score



All ground-truth dog boxes



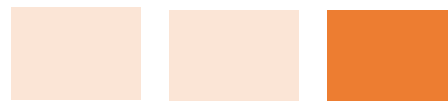
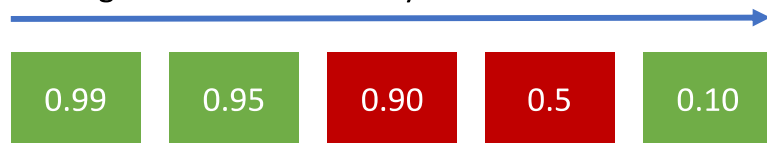
Evaluating Object Detectors

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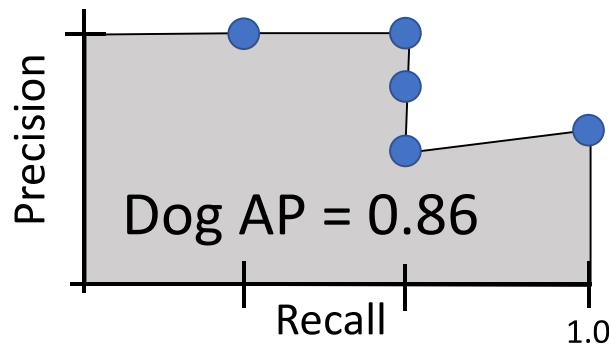
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How to get AP = 1.0: Hit all GT boxes with IoU > 0.5, and have no “false positive” detections ranked above any “true positives”

All dog detections sorted by score



All ground-truth dog boxes



Evaluating Object Detectors

Mean Average Precision (mAP):

1. Run object detector on all test images (with NMS)
2. For each category, compute Average Precision (AP) = area under Precision vs Recall Curve
 1. For each detection (highest score to lowest score)
 1. If it matches some GT box with IoU > 0.5, mark it as positive and eliminate the GT
 2. Otherwise mark it as negative
 3. Plot a point on PR Curve
3. Mean Average Precision (mAP) = average of AP for each category

Car AP = 0.65

Cat AP = 0.80

Dog AP = 0.86

mAP@0.5 = 0.77

Evaluating Object Detectors

Mean Average Precision (mAP):

1. Run object detector on all test images (with NMS)
2. For each category, compute Average Precision (AP) = area under Precision vs Recall Curve
 1. For each detection (highest score to lowest score)
 1. If it matches some GT box with IoU > 0.5, mark it as positive and eliminate the GT
 2. Otherwise mark it as negative
 3. Plot a point on PR Curve
3. Mean Average Precision (mAP) = average of AP for each category
4. For “COCO mAP”: Compute mAP@thresh for each IoU threshold (0.5, 0.55, 0.6, ..., 0.95) and take average

mAP@0.5 = 0.77
mAP@0.55 = 0.71
mAP@0.60 = 0.65
...
mAP@0.95 = 0.2

COCO mAP = 0.4

Today's Agenda



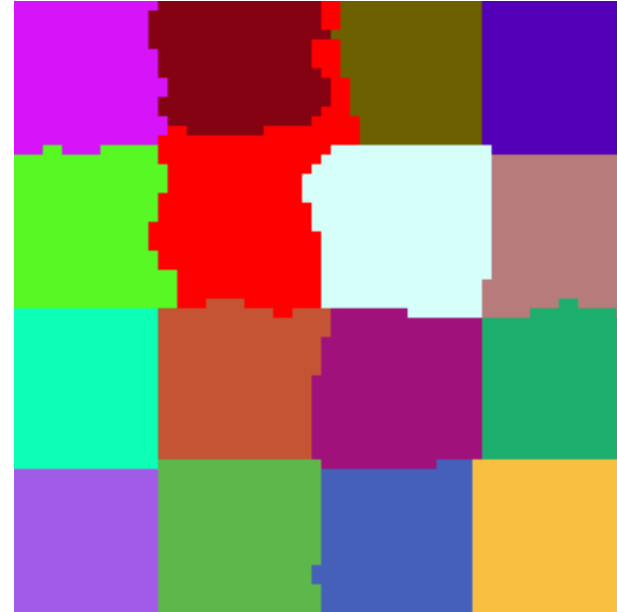
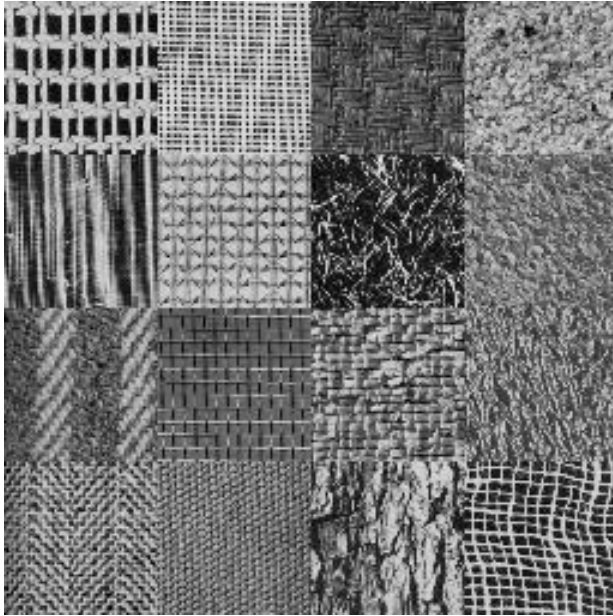
- Image Retrieval
- Object Detection
- **Image Segmentation**

Image Segmentation

- Goal: Partition the image into semantically-meaningful or perceptually-similar regions.



Texture Segmentation



Courtesy Uni Bonn

Image Segmentation

- “Grabcut”
Graphcut-based
Image segmentation
- Integrated into MS Office
e.g. Powerpoint



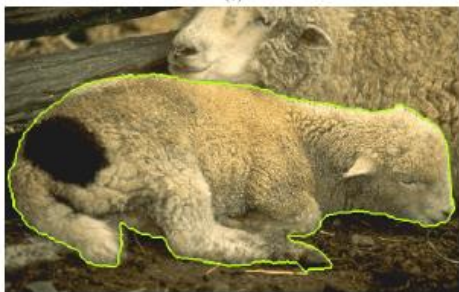
Image Segmentation



(a)



(b)



(c)



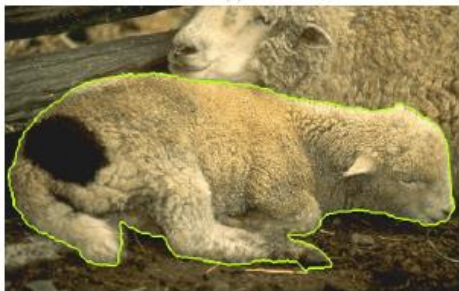
Image Segmentation



(a)



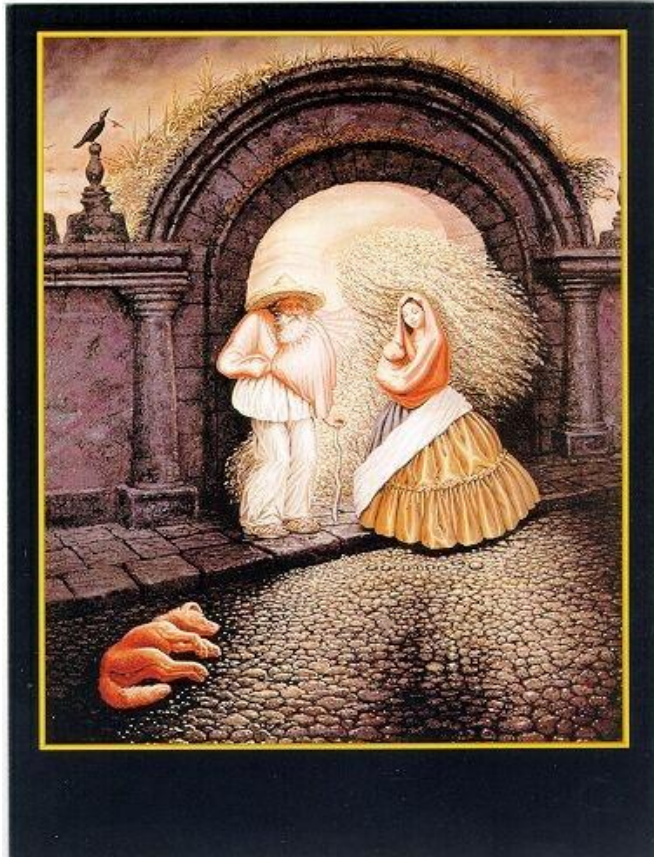
(b)



(c)



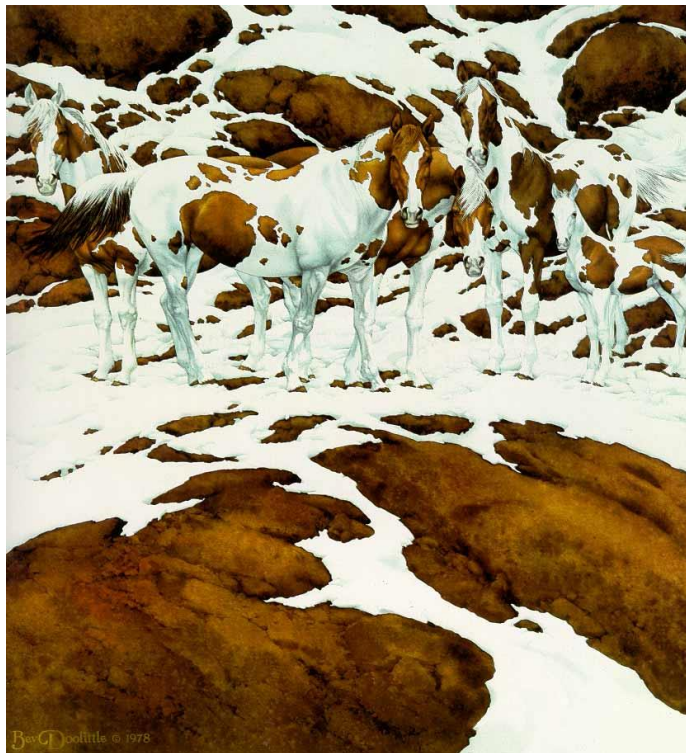
Segment first, or recognise first? (what is the context?)



A portrait of
an old man, or

A couple with
a dog on the street

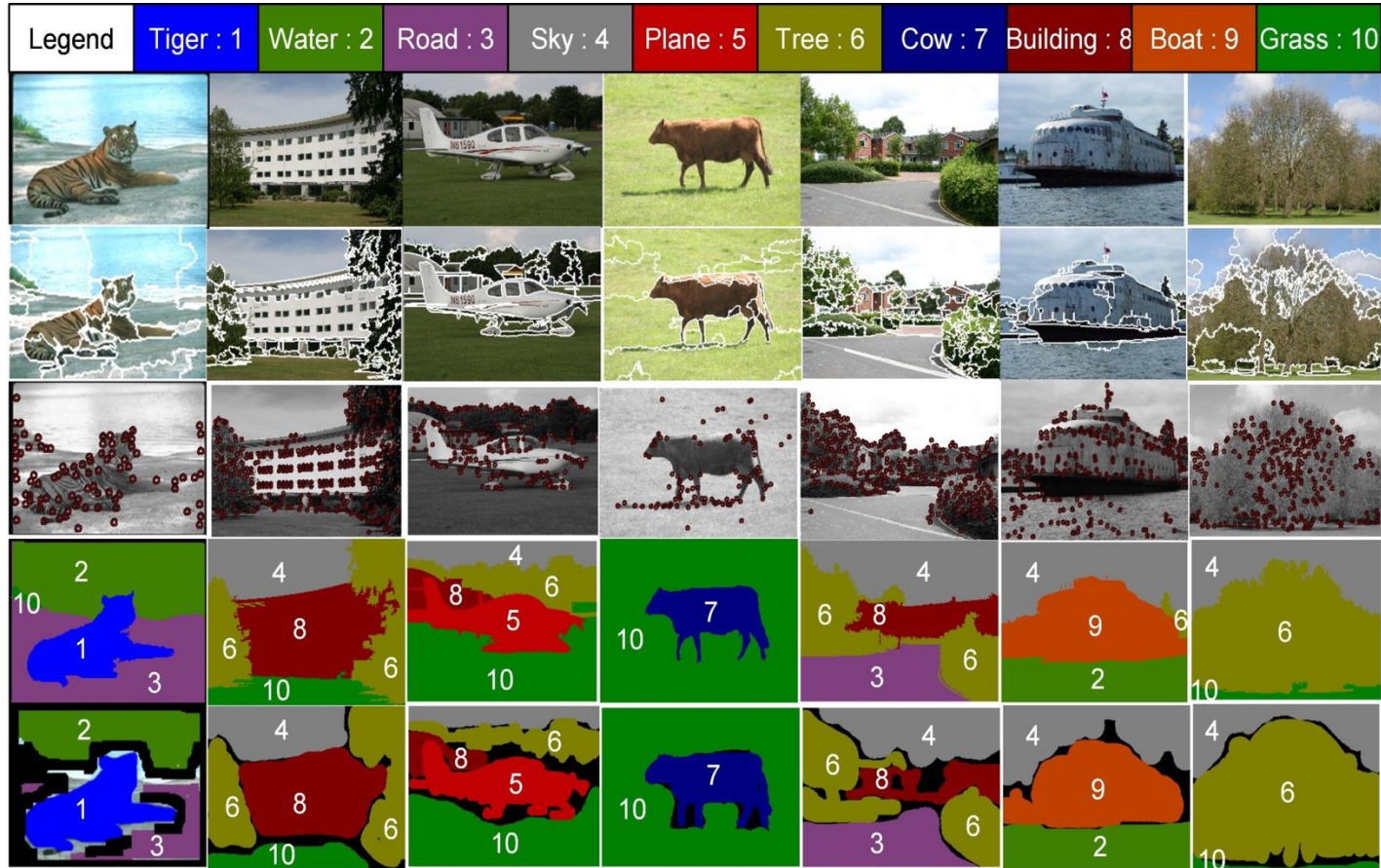
Segment first, or recognise first? (what is the context?)



Snow and rocks,
or

A group of horses
In snow and rocks

Segmentation for Recognition



Today's Agenda

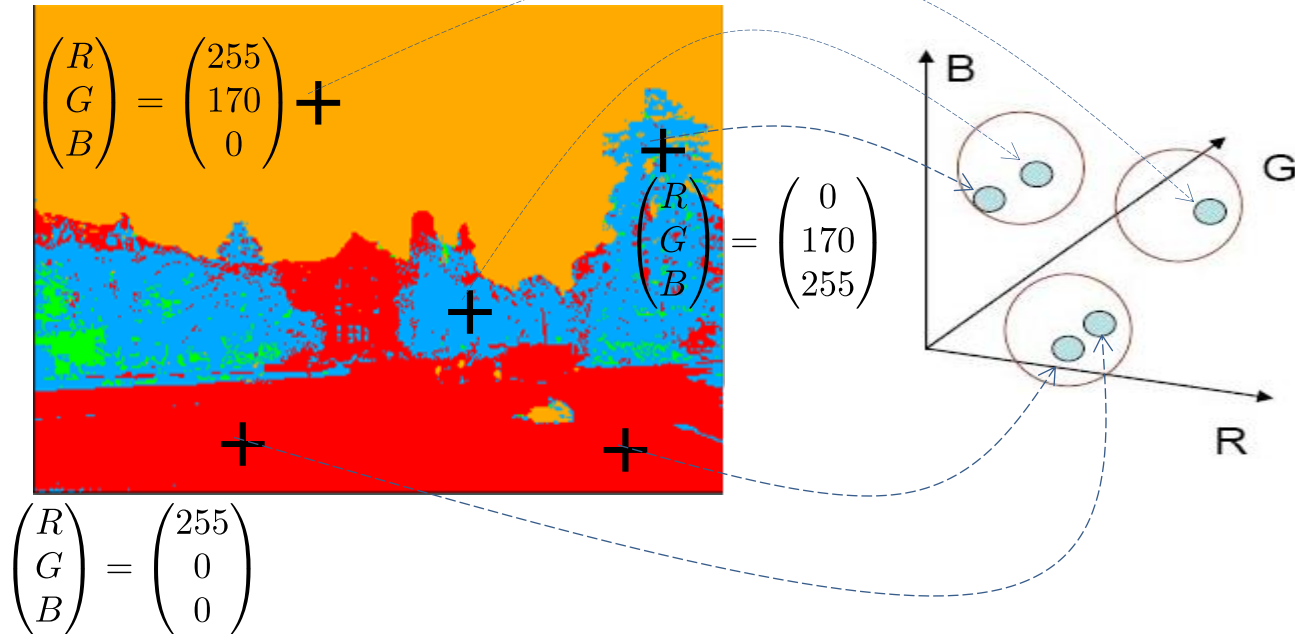
- Image Retrieval
- Object Detection
- **Image Segmentation**
 - K-Means

Segmentation as Clustering

- Low- and mid-level visual cues for segmentation.
- Partition the image pixels into groups with perceptually homogeneous or similar pixel properties (e.g., intensity, colour, texture, or similar spatial location).
- Grouping pixels into perceptually homogeneous regions.

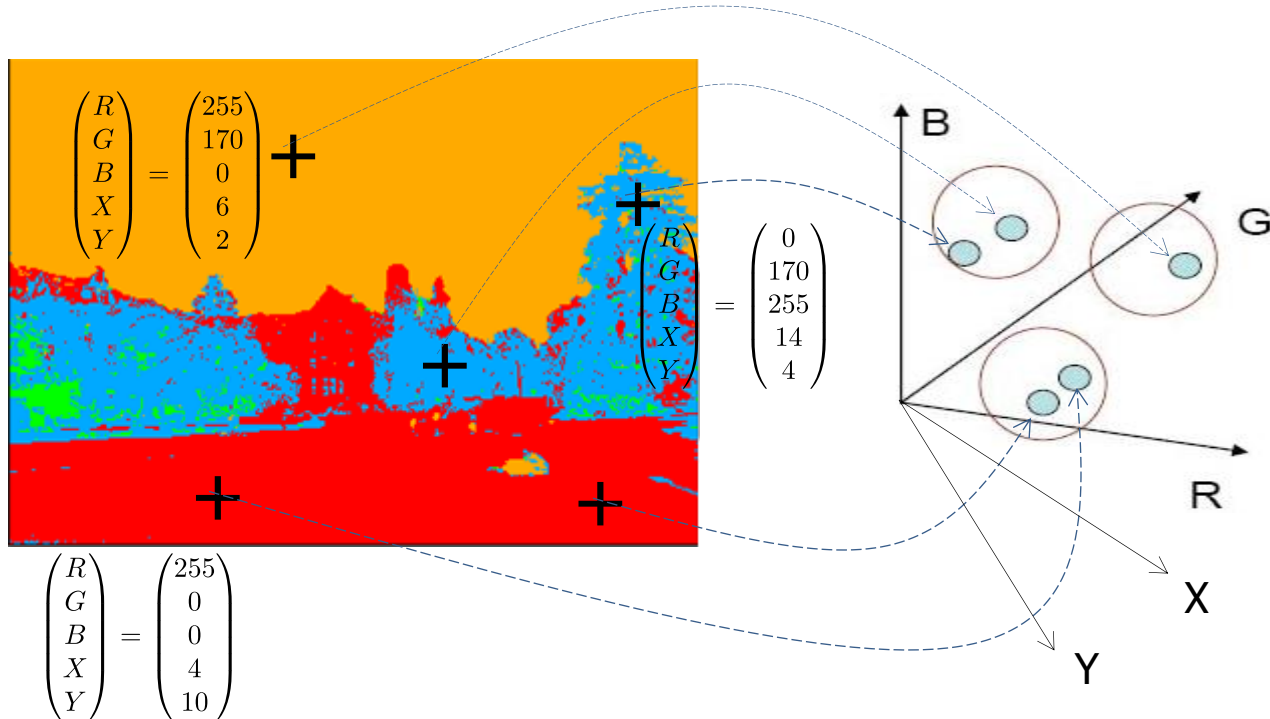
Segmentation as Clustering

- Cluster similar pixels using colour features



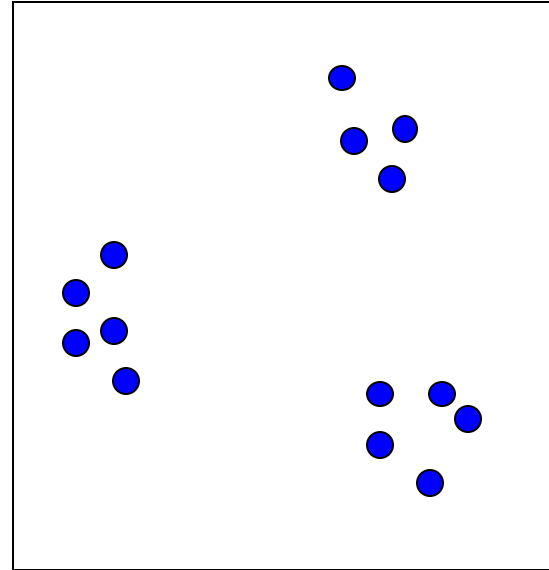
Segmentation as Clustering

- Cluster similar pixels together (RBG+ XY)



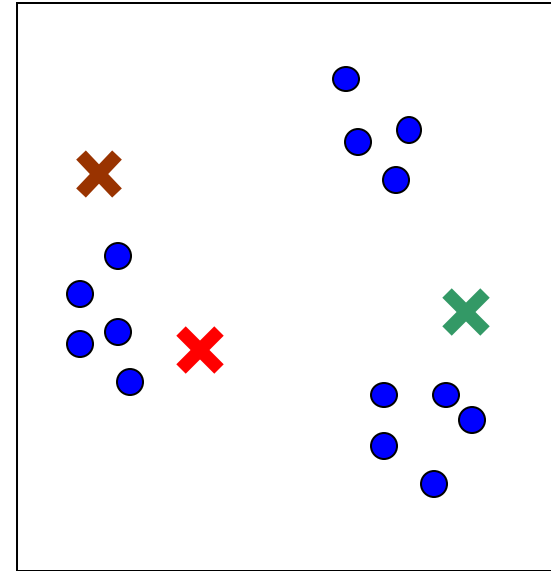
K-Means

- “Guess” the number of clusters K before start (e.g. $K=3$)



K-Means

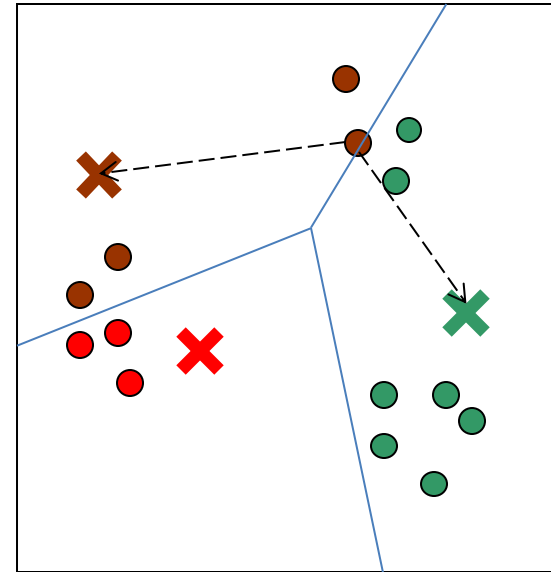
- Start with 3 *random* positions of the cluster centers.



Iteration = 0

K-Means

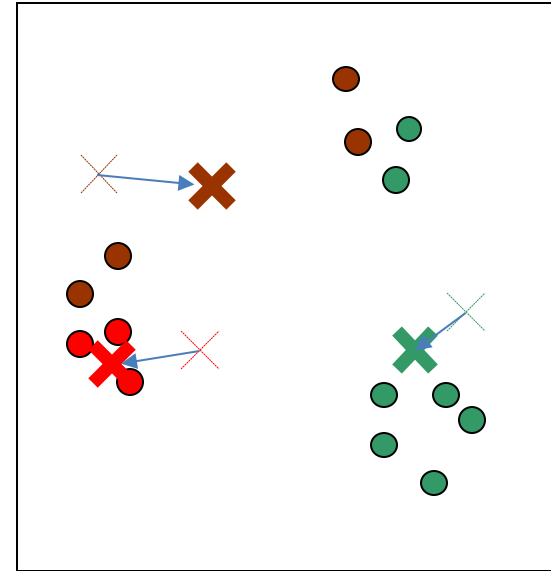
- Start with 3 random positions of the cluster centers.
- By computing the distance, *assign* each data point to the closest center.



Iteration = 1

K-Means

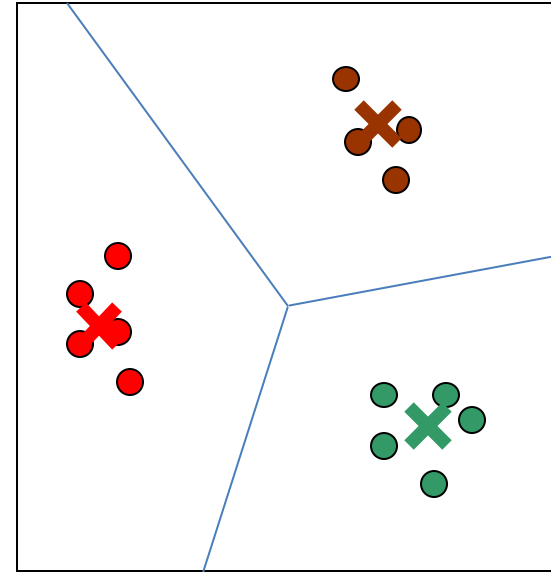
- Start with 3 random positions of the cluster centers.
- By computing the distance, *assign* each data point to the closest center.
- *Re-compute* the centers after the assignments.



Iteration = 1

K-Means

- Start with 3 random positions of the cluster centers.
- By computing the distance, *assign* each data point to the closest center.
- Re-compute the centers after the assignments.
- Iterate until no points are reassigned.



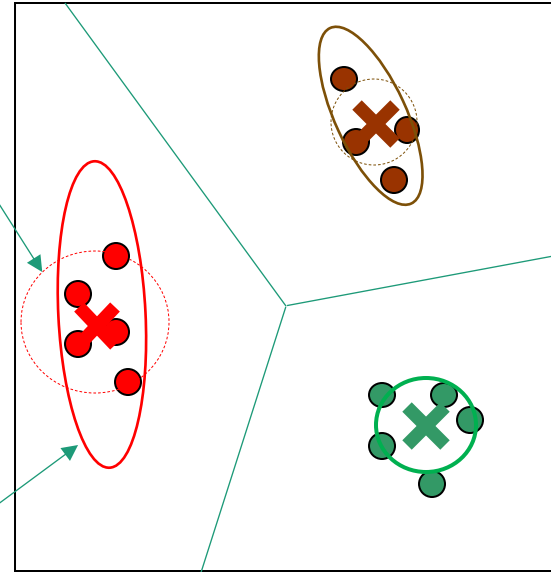
Iteration = 3

K-Means

- Question: What have we minimized?
- We have selected the means and membership which minimize the sum of squared distances to the center means (centroids).
- The sum of squared distances to the means is actually the variance of the cluster

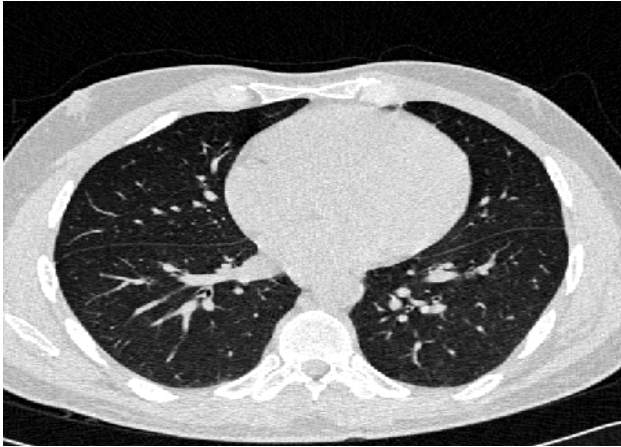
K-Means

- Spherical variance is used
- Q) What about the elongated variances?
 - Called Gaussian Mixture Model based clustering

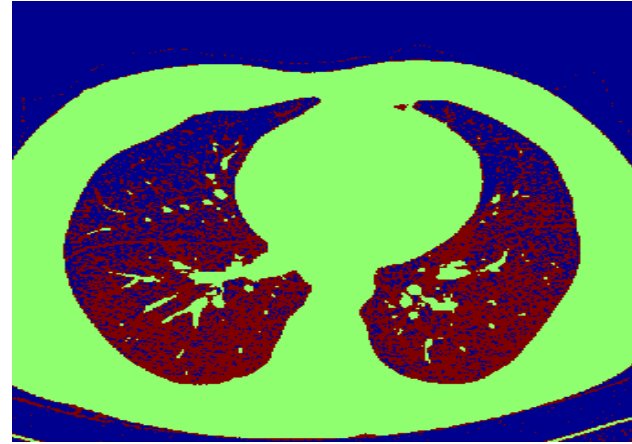


K-Means

1. Select a value of K
2. Select a feature vector for every pixel (color, texture, position, or combination of these etc.)
3. Define a similarity measure between feature vectors (Usually Euclidean distance)
4. Apply the K-means algorithm to all the feature vectors



Input image (I)



Three-cluster image using the
gray levels of *input image*

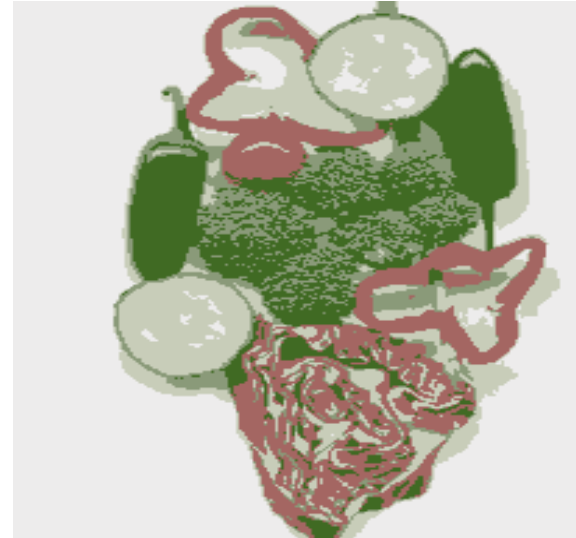
Matlab code:

```
I = double(imread('...'));
```

```
J = reshape(kmeans(I(:),3),size(I)); % need statistics toolbox
```



Color Image



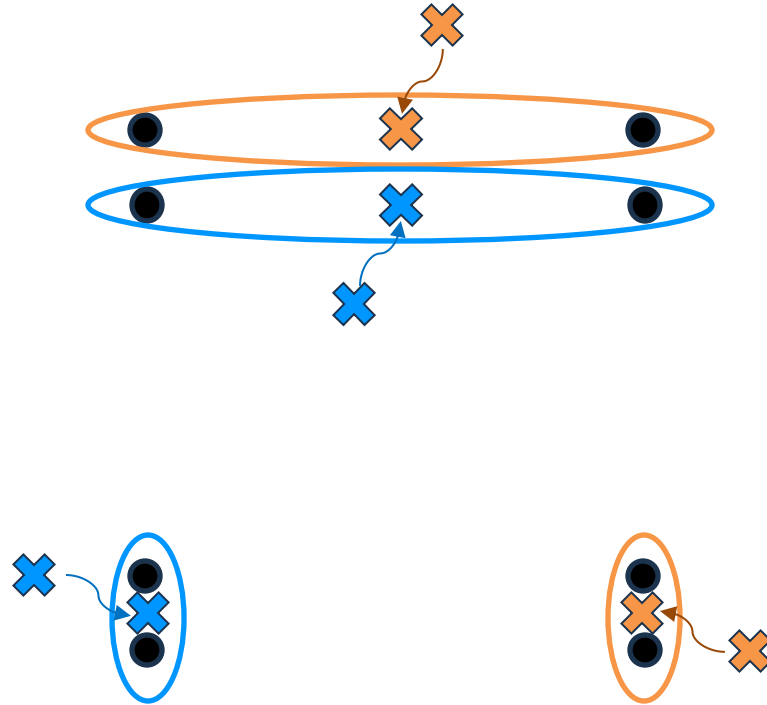
Segmented on color

K-means clustering used for reducing the number of colors (e.g. $k=4$)

K-Means

Local Minima

- Sensitive to initialization
→ sub-optimal solution



Optimal solution

K-Means

Pros:

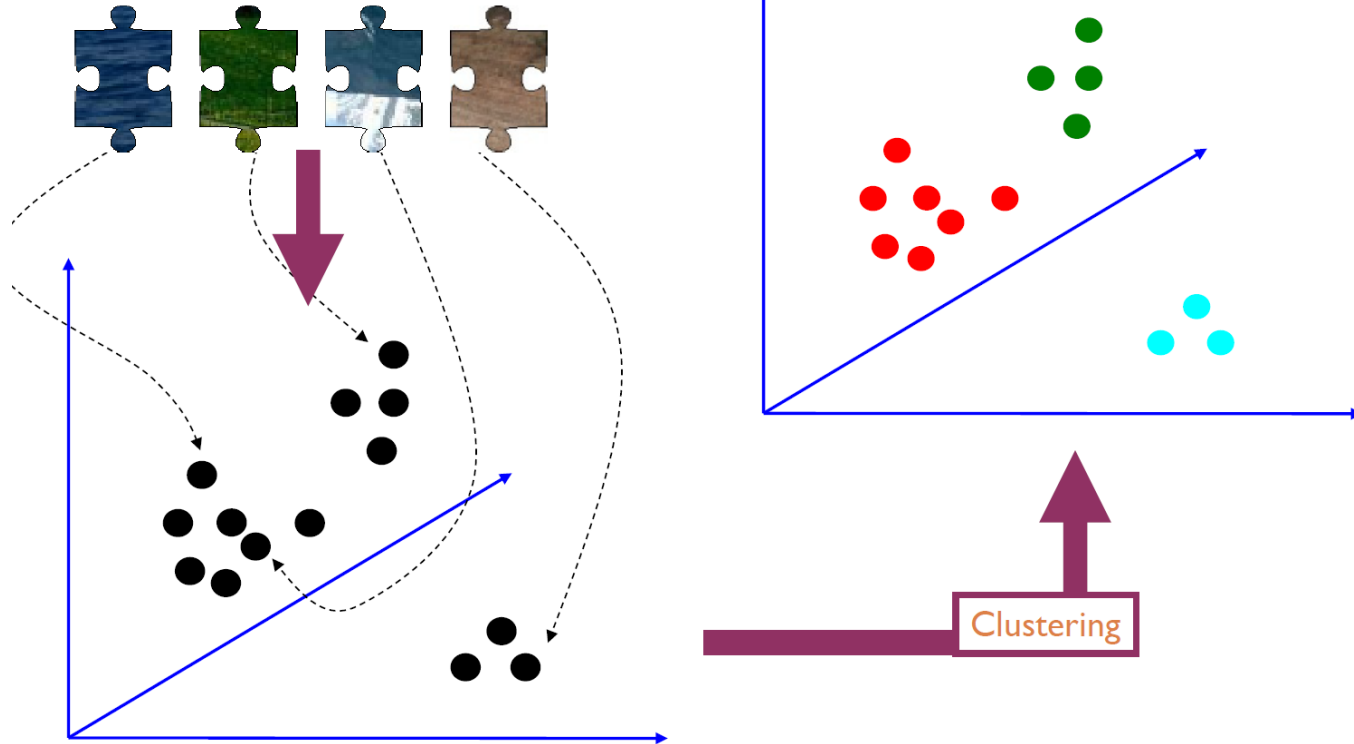
- Finds cluster centers that minimize variance (good representation of data)
- Simple to implement, widespread applications.

Cons:

- All clusters have spherical distribution (same to all directions or isotropic)
- Prone to local minima
- Need to choose K
- Can be very slow: each iteration is $O(KN)$ for N-dimensional points

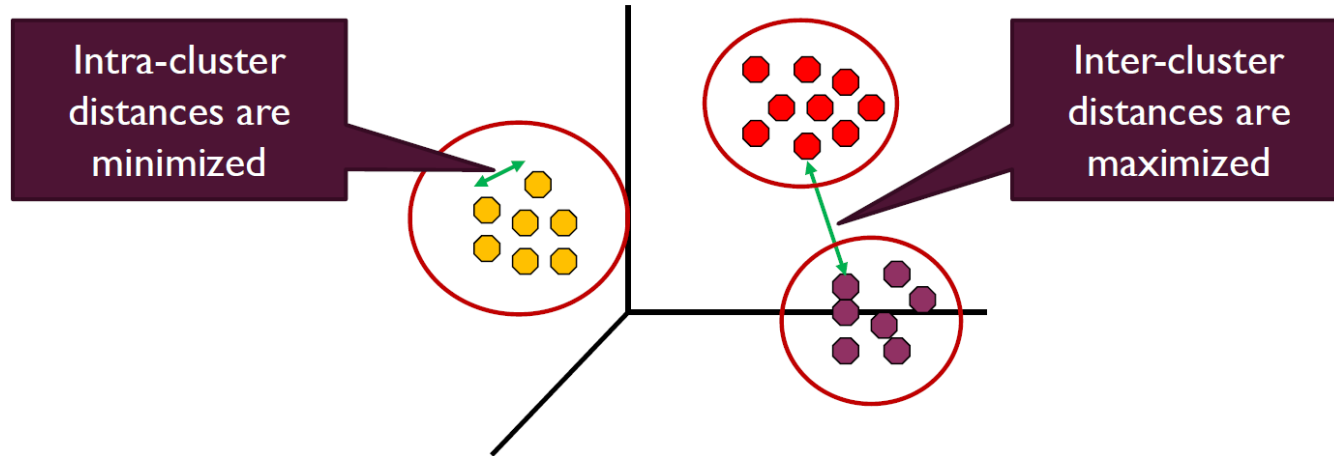
Reminder: Feature Clustering for BoW

CLUSTERING FOR BOW



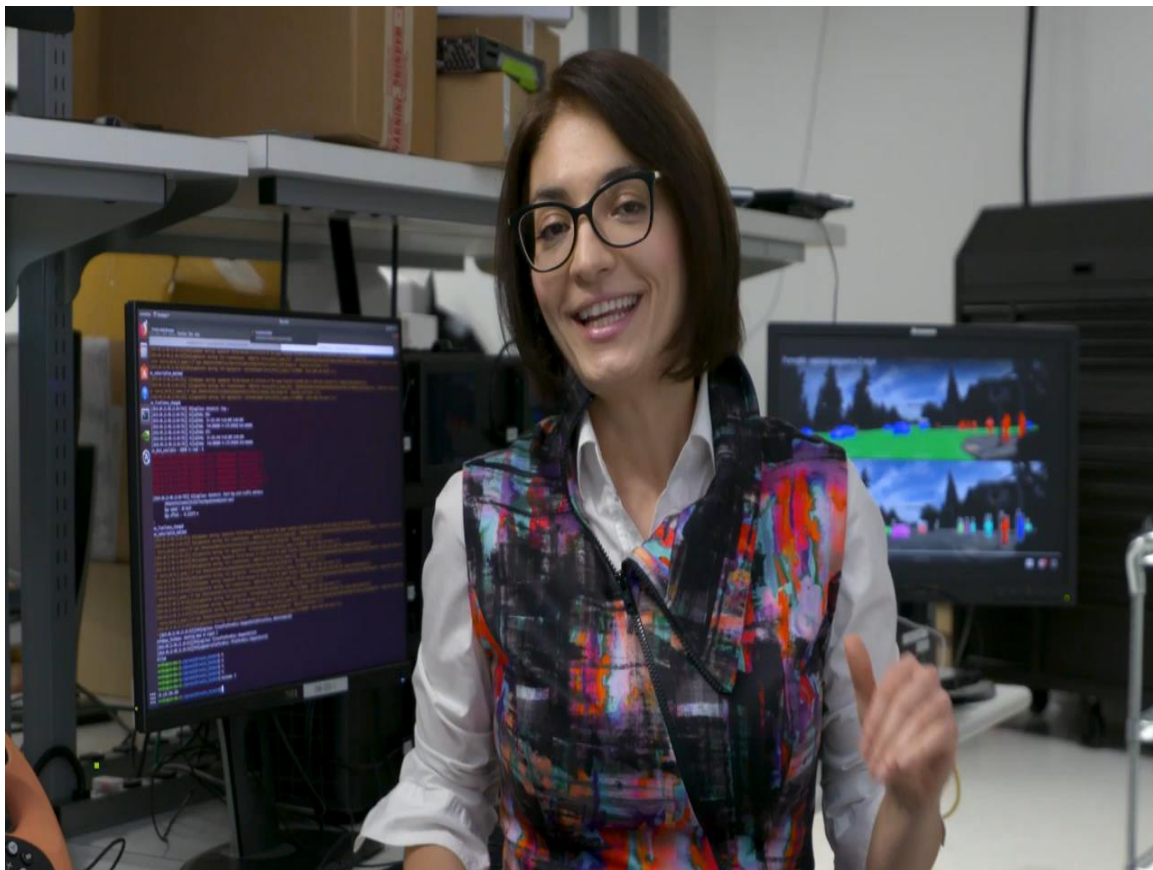
Reminder: Feature Clustering for BoW

Find clusters of similar descriptors





Panoptic Segmentation (Semantic+Instance)



Questions?

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